ScienceDirect

Journal of Manufacturing Processes

Date: 26 December 2024

Show more

Research article Full text access Get rights and content

Effect of shielding gas on keyhole stability and pores in laser-CMT hybrid welding of ultra-high strength steel

Yuhui Zhang [a b], Cong Chen [a b], Changjian Wang [c], Ran Xiong [c], Ke Zhang [a b] Show more

View PDF Download full issue Cite Add to Mendeley Share

10.1016/j.jmapro.2024.11.042

Article Recommended articles Cited by (5) Metrics

Highlights

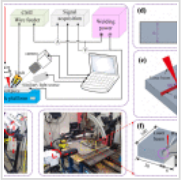
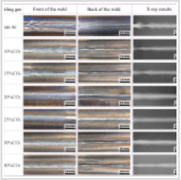
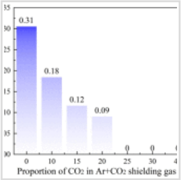
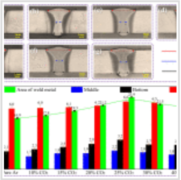
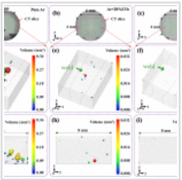
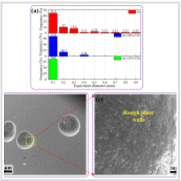
Outline

- Laser-CMT hybrid welding was successfully employed in aerospace ultra-high strength steel welding.
- A free porosity and sound welded joint of UHSS was achieved by adjusting the CO2 content to 25%.
- Droplet transfer and molten pool flow behaviors were observed through high-speed images and electrical signals.
- Evolution of the keyhole internal shape was summarized with different shielding gases.
- Keyhole stability and pore suppression mechanisms of the laser-CMT hybrid welding were revealed.

Abstract

Laser-CMT hybrid welding (LCHW) is a highly efficient and high-quality welding technique, showing promise for welding ultra-high strength steel (UHSS) in the aerospace industry. However, the presence of pores significantly limits its application in this field. It has been observed that using an argon-rich gas can effectively stabilize the keyhole and prevent the formation of pores. Therefore, this study aims to investigate the effect of different shielding gases on keyhole stability and porosity. The weld profile, pores, electrical signal, droplet transfer, and molten pool internal flow were carefully characterized and compared with various shielding gases. The results show that increasing the CO_2 content gradually reduces weld porosity. The optimal weld formation and porosity were achieved at a CO_2 content of 25%. With the CO_2 content increases, the droplets shift from off-axis upwards to downwards. This shift increases the distance between the droplet and the keyhole, reducing the time for droplet growth and promoting droplet transfer. Furthermore, the surface tension pressure gradually decreases from 3.86 Kpa to 1.1 Kpa, and at Ar+25% CO_2, itpromote the opening of keyhole and further suppress the formation of the protrusion on the keyhole rear wall. In addition, when the CO_2 content reaches 25%, the top metal of the molten pool changes from

Figures (18)

Show 12 more figures

Tables (3)

counterclockwise flow to clockwise flow, reducing the likelihood of keyhole collapse. These effects stabilize the keyhole and suppress the formation of pores.

Keywords

Ultra-high strength steel; Shielding gases; Laser-CMT hybrid welding; Pore; Laser keyhole stability

Previous article in this issue

Next article in this issue

1. Introduction

Ultra-high strength steel (UHSS) is widely used in aerospace, defense, and automotive industries, as well as other important fields, due to its ultra-high strength and toughness [[1], [2], [3], [4]]. 30Si2MnCrMoVE (known as D406A) steel is a typical low alloy UHSS with a tensile strength >1600 MPa and an elongation higher than 10% in the quenching and tempering states [5]. Welding is a crucial processing technique for these materials. Nevertheless, the high carbon equivalent results in poor weldability, increasing the likelihood of defects like cracks [6]. Currently, due to the stable arc, beautiful forming, and mature welding process that is not prone to defects, the welding of D406A is still mainly done with tungsten inert gas (TIG) welding [[7], [8], [9]]. However, the high welding heat input and slow welding speed leads to a significant welding distortion and low production efficiency of the welded joint [10]. Therefore, it is imperative to develop a new high-quality and high-efficiency welding method for D406A UHSS.

Laser-arc hybrid welding combines the benefits of both laser and arc welding, ensuring high efficiency and quality while also demonstrating strong adaptability to gaps [11]. Laser can not only achieve deep penetration but also stabilize the welding arc [12,13]. The arc can modulate the microstructure and properties of the weld by altering the composition of the welding wire and the wire feed rate, realizing an effect of 1+1>2 [14]. More importantly, cold metal transfer (CMT), as a special gas metal arc welding (GMAW) welding method, has excellent arc stability and low heat input, reducing the tendency of welding distortion [15]. Combining it with lasers may produce better results. Therefore, laser-CMT hybrid welding (LCHW) has attracted more and more attention in recent years. Frostevarg et al. [16] argued the advantages and disadvantages of laser-arc hybrid welding with three different arc modes (standard, pulsed and CMT). It was found that the CMT mode showed advantages of higher stability, reduced undercut and decreased weld and heat affected zone (HAZ) widths. Lei et al. [17] explained the impact of laser thermal effects on the droplet transfer behavior of LCHW. The results showed that the laser enhanced the melting of the filler wire, promoted the spreading of the molten pool, increased the frequency of droplet transfer, and improved the weld formation. Moreover, Li et al. [18] found that using the CMT arc mode in laser-arc hybrid welding resulted in a spatter-free process, leading to a uniform structure, fine grain size, and improved mechanical properties. In view of the aforementioned advantages, LCHW has a high application prospect in the field of aerospace UHSS welding.

However, the droplet transfer behavior during hybrid welding will disturb the stability of the laser keyhole, leading to the keyhole collapse and the formation of pores [19]. Studies have shown that the welding process had a significant impact on the stability of the laser keyhole, and unstable keyholes can produce porosity defects [20]. Additionally, due to the high viscosity coefficient of the liquid metal in the D406A [9,21], bubbles formed during the welding process were difficult to escape [22], ultimately resulting in porosity defects in the weld metal. Nevertheless, pores reduce the mechanical properties of the product [23] and cannot meet the requirements in complex environments.

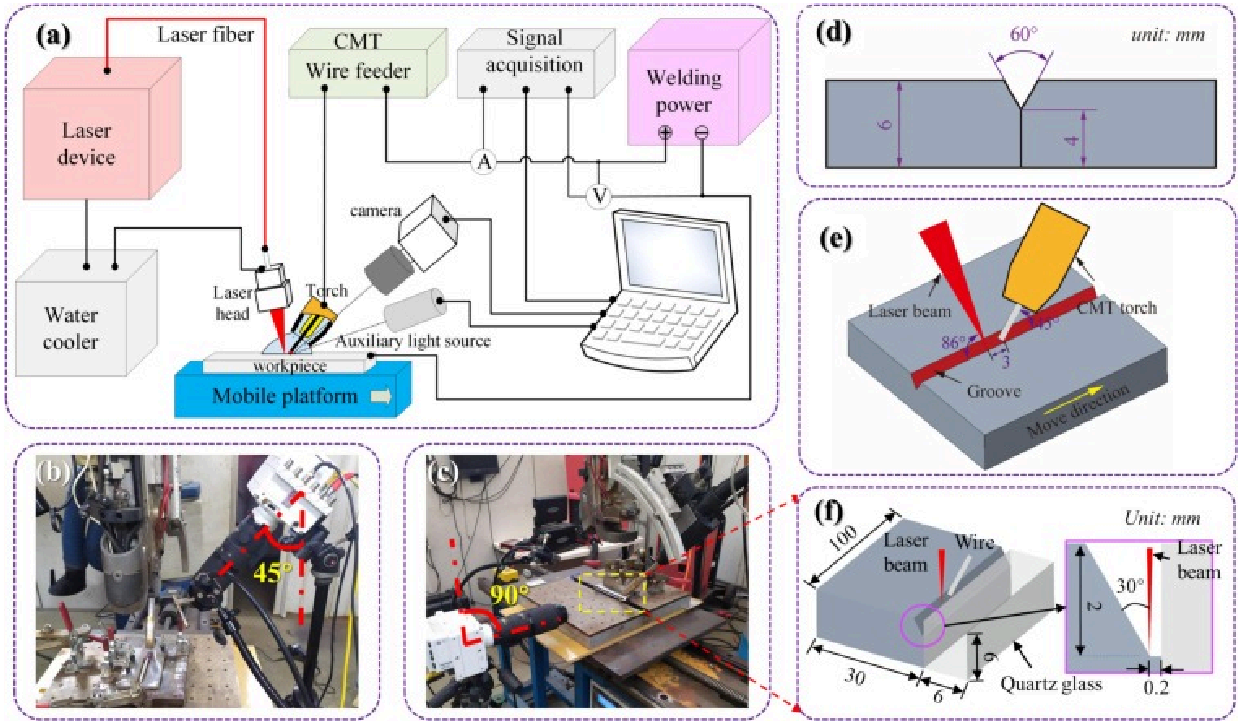
Our previous experiments have found that the shielding gas had a significant effect on weld morphology and porosity in D406A. Relevant studies have also confirmed that different shielding gas plays a vital role in enhancing the stability of hybrid welding and eliminating porosity [[24], [25], [26], [27]]. Anna Fellman et al. [28] used Ar, He and CO_2 in different proportions as shielding gases for laser arc hybrid welding. It was found that good welding results could be obtained when the shielding gases were Ar+40%–50% He and Ar+2%–5% CO_2. Meanwhile, studies have shown [29] that during laser arc hybrid welding, the keyhole opening gradually stabilizes as the He content in the Ar+He shielding gas increases. When the volume ratio of He exceeds 50%, the shielding effect of laser-induced plasma is suppressed, so the keyhole remains open during the welding process. Zhang et al. [30] examined the impact of various argon-rich shielding gases on the stability of laser-MAG hybrid welding of high-strength steel. The findings revealed that as the CO_2 content increased, the droplet transfer mode transitioned from an unstable granular mode to a stable spray mode, subsequently evolving into a repulsive mode. However, when pure CO_2 was employed, the metal transfer mode was characterized by an unstable repulsive transition, leading to considerable welding spatter. Pan et al. [31] and Wahba et al. [32] utilized Ar+20%CO_2 and 100%CO_2 shielding gases to weld high-strength steel by laser-MAG hybrid welding, and found that compared with Ar+20%CO_2 shielding, the porosity rate in the weld was significantly lower for pure CO_2 shielding. Moreover, Zhao et al. [33] elucidated the influence of oxygen on the porosity of steel in laser-MAG hybrid welding. It was found that adding a small amount of oxygen to the shielding gas can stabilize the laser keyhole and inhibit the formation of pores. Mazar Atabaki et al. [22] used two kinds of shielding gas (92% Argon-8% CO and 65% Ar-35% He) to study the effect of shielding gas on porosity in laser-arc hybrid welding of ultra-high strength steel. The results indicated that when using 92% Argon-8% CO as the shielding gas, the number and size of pores decreased by 6.75 times and 12.5 times, respectively. From the studies mentioned above, it is evident that a certain amount of oxidizing shielding gas has a positive impact on welding quality.

However, up to now, few scholars have studied the LCHW process under different shielding gases. Moreover, there is a lack of in-depth exploration into the effect of keyhole stability and the pore suppression mechanism during various phases of CMT when using different shielding gases on UHSS. In this study, we investigate the impact of shielding gases on keyhole stability and pore suppression in LCHW of D406A UHSS. Our findings reveal the mechanisms of keyhole stability and pore suppression, and the strategy for eliminating pores is also proposed. Firstly, the surface and internal quality of post-weld samples under different shielding gases were characterized to confirm the distribution and type of the pore. Secondly, by analyzing the electrical signals and high-speed images during the welding process, the behavior of CMT arc and droplet transfer under different proportions of CO_2 was discussed. Thirdly, the sandwich experiment method was employed to observe and analyze the flow behaviors of liquid metal and the shape evolution of the keyhole in the molten pool under different shielding gases. Finally, based on the above experimental phenomena and analysis, the mechanism of the droplet transfer process, keyhole evolution, and pore suppression process during LCHW under different shielding gases were explored through force balance theory and keyhole stability condition.

2. Materials and methods

2.1. Materials and welding equipment

The schematic diagram of the LCHW experimental system is shown in Fig. 1(a). An IPG fiber laser (YLS-10000) with a wavelength of 1070 nm was used as the laser heat source. The laser beam was focused on a spot of 0.72 mm in diameter through a focal length of 400 mm. The Fronius (Trans Pulse Synergic 5000R) served as the welding power supply, equipped with a VR-1550 wire feeder for CMT during the welding process. The laser head and CMT torch were fixed together using adjustable mechanical devices. To ensure clear observation of droplet transfer and molten pool dynamics during welding, the welding head remained stationary, while a computerized numerical control (CNC) mobile platform facilitated a smooth welding process.



Download: Download high-res image (664KB)

Download: Download full-size image

Fig. 1. Laser-CMT hybrid welding system: (a) schematic of welding system, (b) high-speed camera with a vertical downward view, (c) high-speed camera with a vertical horizontal view, (d) schematic diagram of weld groove, (e) magnification of the welding position, (f) quartz glass placement and laser incident position.

In this study, the base material (BM) used was D406A UHSS in an annealed state. The filler wire used was H10SiMnCrNiMoV with a diameter of 1.2 mm. Their nominal chemical compositions of both materials are listed in Table 1. The specimens were machined to a size of 110 mm (length) × 50 mm (width) × 6 mm (thickness) with a 30°single Y groove and 4 mm blunt, as shown in Fig. 1(d). Prior to welding, the plate's surface was cleaned of oxides and oil using mechanical polishing and alcohol, and then secured in position with a fixture. The laser head and CMT torch were adjusted to incline at 86° and 45°

respectively from the horizontal plane. The wire was extended 14mm and maintained 3mm from the laser focal point, with the wire extended into the groove for welding, as illustrated in Fig. 1(e). The specific experimental parameters for the welding process are detailed in Table 2. The primary focus of the study was to investigate the impact of different shielding atmospheres (pure Ar, Ar+10%CO_2, Ar+15%CO_2, Ar+20%CO_2, Ar+25%CO_2, Ar+30%CO_2, and Ar+40%CO_2) at a flow rate of 20L/min on weld formation, droplet transfer, and molten pool flow behavior.

Table 1. Nominal compositions (wt%) of base metal and filler wire.

Material	C	Mn	Si	Ni	Cr	Mo	V	Fe
Base metal	0.28–0.33	0.7–1.0	1.4–1.7	0.25	1.0–1.3	0.4–0.55	0.08–0.15	Bal.
Filler wire	0.08–0.12	0.9–1.1	1.1–1.3	1.8–2.0	1.4–1.6	0.4–0.6	0.05–0.15	Bal.

Table 2. Process parameters of laser-CMT hybrid welding.

Parameters	Value
Laser power $\boldsymbol{P}$ (W)	4300
Welding speed $\boldsymbol{V_w}$ (m/min)	0.9
Wire feeding rate $\boldsymbol{V_f}$ (m/min)	6
Defocus amount $\boldsymbol{d}$ (mm)	0
Distance between laser and wire $\boldsymbol{D_{la}}$ (mm)	3
CO_2 content in shielding gas	**0, 10, 15, 20, 25, 30, 40**
Wire extension $\boldsymbol{l}$ (mm)	14
Shielding gas flow rate $\boldsymbol{R}$ (L/min)	20

2.2. Experimental and analysis methods

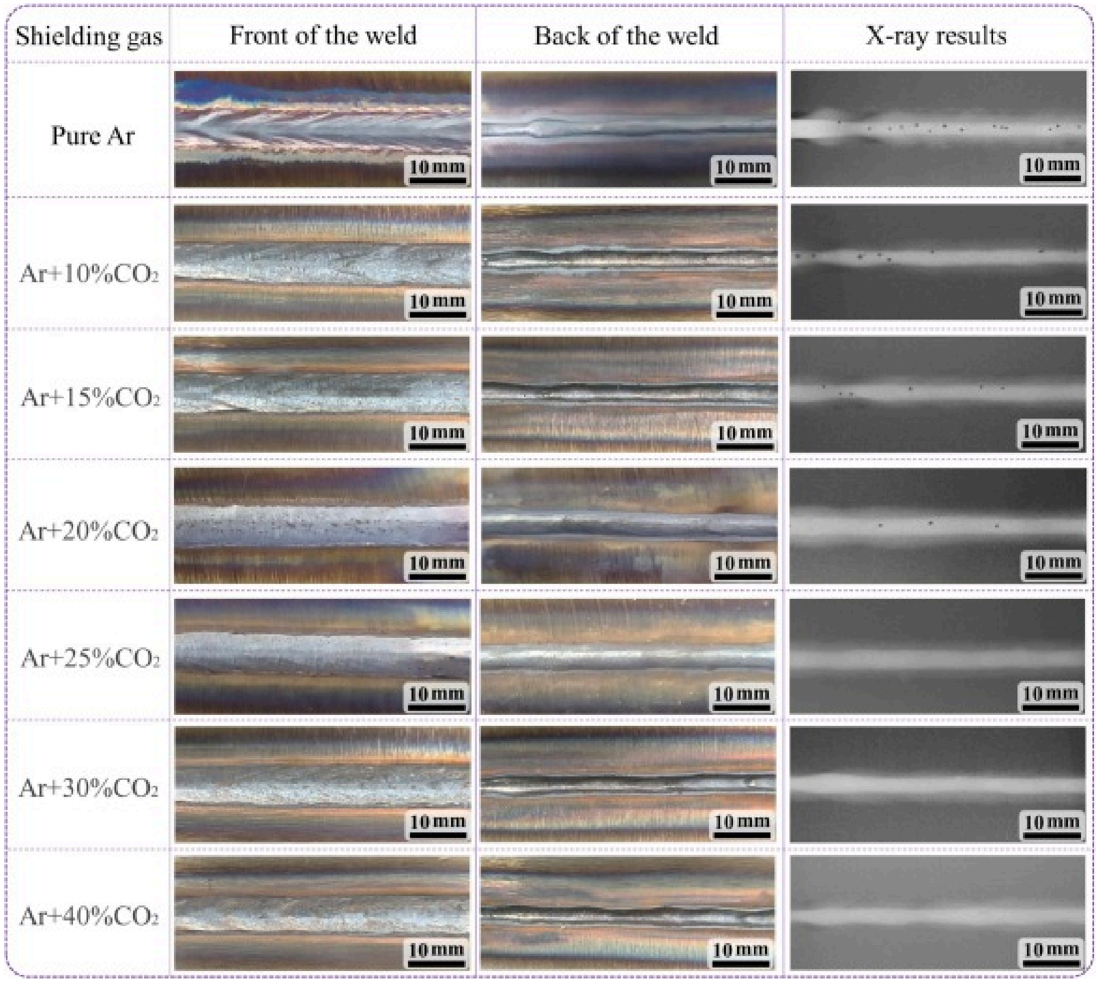
During the welding process, the droplet transfer process and molten pool flow behavior were visualized by a Photron Fastcam SA4 high-speed camera system. The camera was positioned at a 45° angle in the vertical downward view and at a 90° angle in the vertical horizontal view using transparent quartz glass (known as the sandwich method), respectively, as depicted in Fig. 1(b) and Fig. 1(c). Further information about the placement of the quartz glass ($100\text{mm} \times 6\text{mm} \times 6\text{mm}$) and the laser incidence position can be found in Fig. 1(f). The high-speed images were taken at 5000 frames/s. Real-time welding current and voltage were measured and recorded using Hall sensors and acquisition cards, with a measurement frequency of 20 KHz, as shown in Fig. 1(a).

After welding, the plates were first subjected to X-ray detection, followed by the cutting of metallographic specimens. These specimens were then ground, polished, and etched with a 4% nitric acid-alcohol solution to observe the morphology and porosity of the weld joint. Subsequently, the weld and porosity were analyzed using an optical microscope (OM) and Image-J software, respectively. To accurately determine the shape and location of the pores, the X-ray computerized tomography (XCT) was used by Xradia 520 Versa with a voltage of 160 kV. Furthermore, a three-dimensional (3D) scan was conducted on the weld sample with a spatial resolution of 7.6 μm/pixel. After polishing and ultrasonic cleaning, the specimen with pores was observed using a scanning electron microscope (SEM) to assess the pore morphology. The Image-J software was also used to measure the long and short axis diameters of the droplets across 10 cycles and calculate the average value to determine the droplet equivalent diameter. The frequency of droplet transfer was determined by counting the number of droplets occurring within one second under high-speed camera observation.

3. Results and discussion

3.1. Optimization of the shielding gas for weld seam

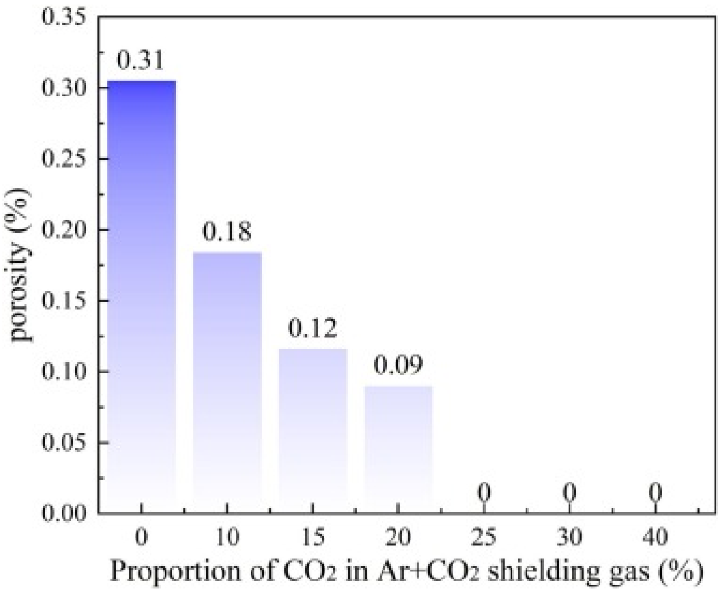
Fig. 2 illustrates the surface morphology of the weld and corresponding X-ray detection results under different CO_2 contents. The results show that, while all samples achieved effective connections, there are significant differences in the weld surface morphology depending on the shielding gas used with the same welding parameters. Under the pure Ar condition, the front side of the weld had a noticeable concavity, with white regions on either side and a zigzag bulge near the fusion boundary. The back side of the weld was narrower and X-ray detection revealed several pores near the center. As the CO_2 content increased, the weld surface became more uniform. Particularly, when the CO_2 content increased to 25%, both the front and back sides of the weld had a smooth and even morphology. However, at a 30% or 40% CO_2 content, a larger number of oxides appeared on the weld surface, causing a degradation in the backside morphology. The porosity statistics from the X-ray detection results are shown in Fig. 3, which clearly shows a decrease in pores as the CO_2 content increases from 0 to 40%, until they completely disappear.



Download: Download high-res image (895KB)

Download: Download full-size image

Fig. 2. The surface morphologies and X-ray results of LCHW under different CO_2 content.

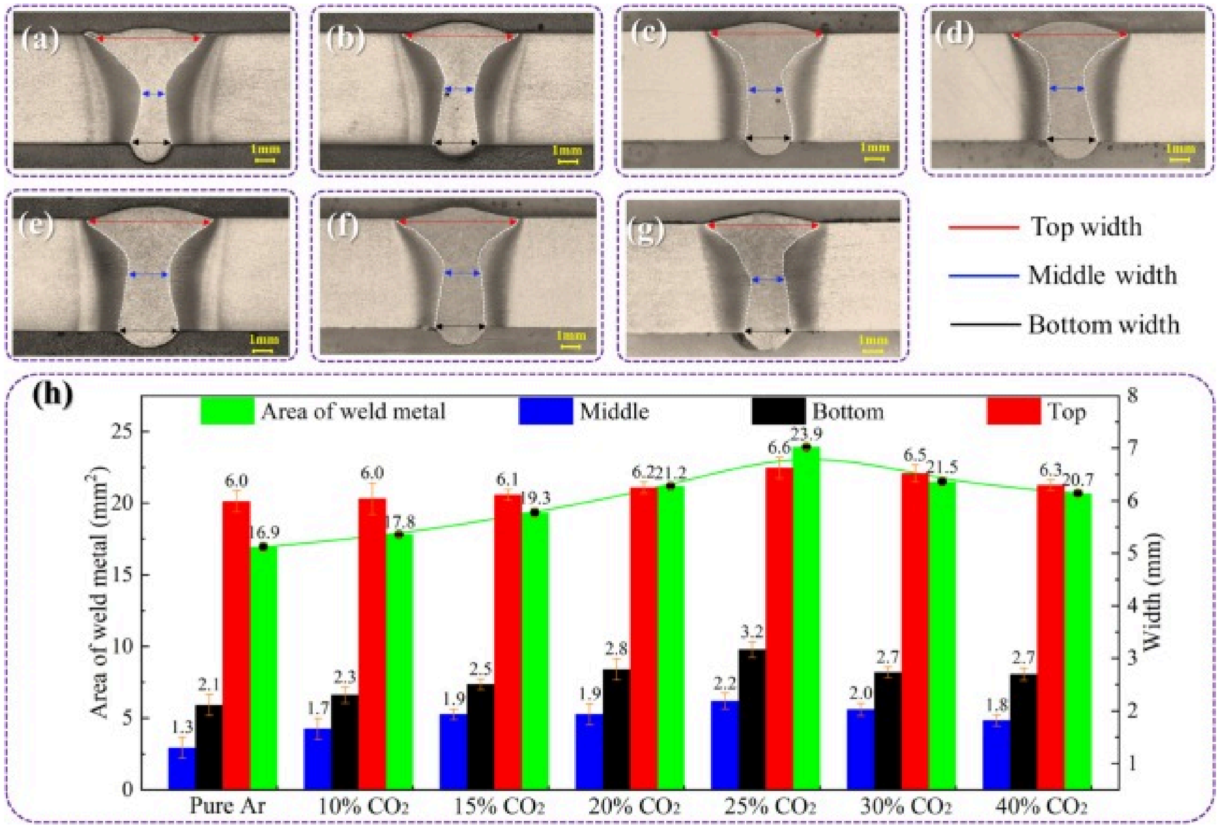


Download: Download high-res image (107KB)

Download: Download full-size image

Fig. 3. Porosity statistics under different CO_2 content.

The cross-sectional macrostructure of LCHW welded joints under different CO_2 contents is shown in Fig. 4(a-g), revealing a pronounced girdle characteristic in all weld cross-sections. Fig. 4(h) shows the distribution of weld width and cross-sectional area characteristics corresponding to different CO_2 contents measured by the Image-J software. It was found that as the CO_2 content increases, the weld width first increases and then decreases. Specifically, when the CO_2 content increases from 0 to 25%, the girdle width of the weld gradually increases, which is beneficial for the escape of pores. However, when the CO_2 content increases to 40%, both the weld width and area exhibit a modest reduction, yet no pores were observed. This phenomenon is attributed to the change in the direction of the surface tension temperature gradient caused by excessive CO_2 addition [34], coupled with arc contraction that enhances the inward flow of molten pool metal [35]. Consequently, the width reduces moderately, but no pores appear.



Download: Download high-res image (656KB)

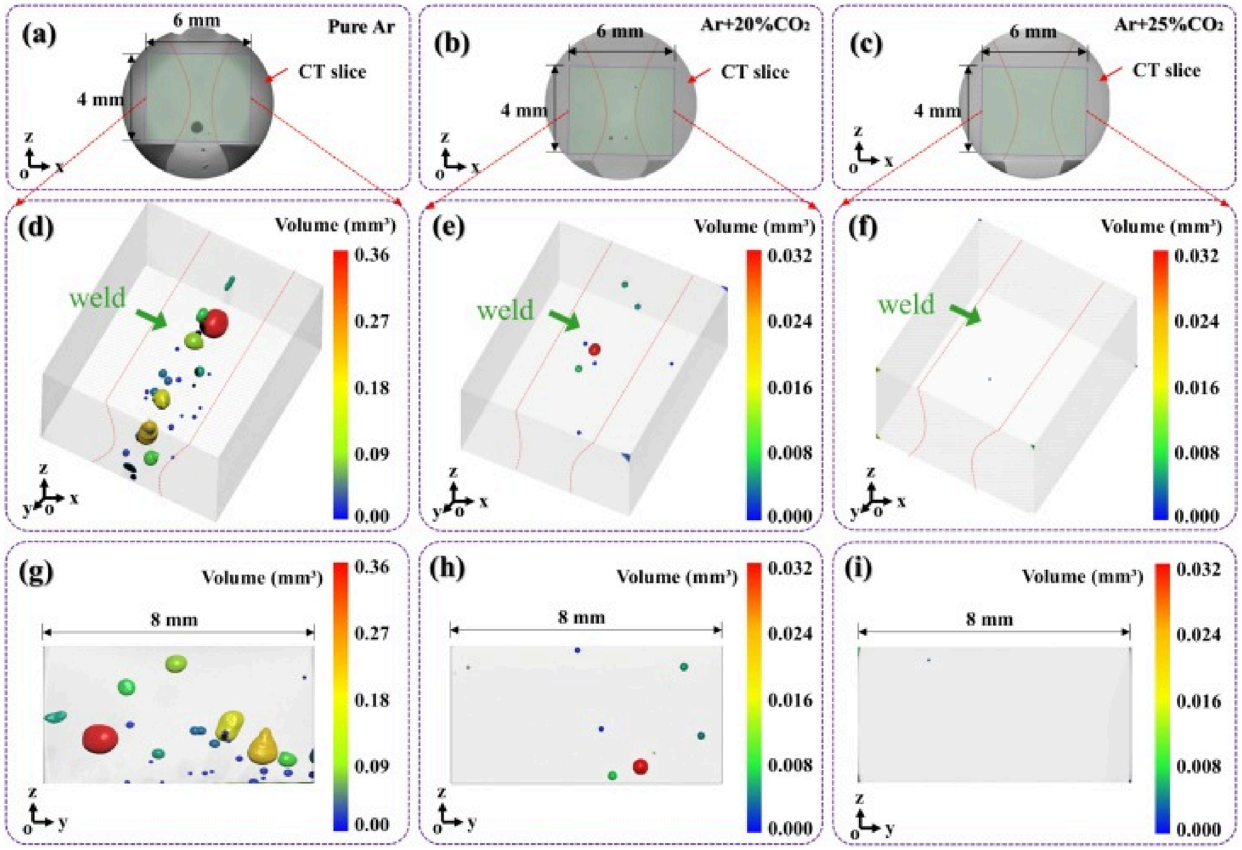
Download: Download full-size image

Fig. 4. The cross-section macrostructure and sizes of LCHW under different CO_2 content: (a) pure Ar, (b) Ar+10%CO_2, (c) Ar+15%CO_2, (d) Ar+20%CO_2, (e) Ar+25%CO_2, (f) Ar+30%CO_2, (g) Ar+40%CO_2 and (h) the distribution of weld metal area and width.

3.2. Pores and electrical signal characteristics

3.2.1. Pores distribution and type

Based on the porosity and macroscopic morphology of the weld, three different shielding gases were selected for analysis: pure Ar, Ar+20%CO_2, and Ar+25%CO_2. In order to accurately characterize the size, distribution, and location of pores within the weld metal, computed tomography (CT) scanning technology was utilized. The 4mm × 6mm welded joint was extracted in the circular CT slice for detailed analysis, as depicted in Fig. 5(a-c). The 3D distribution of porosity can be seen in Fig. 5(d-f). The results show that as the CO_2 content increases, the pores in the weld gradually transition from irregular shapes to spherical shapes, resulting in a decrease in pore count. This can be observed in Fig. 5(g-i), where larger pores are found at the bottom of the weld and smaller pores are randomly distributed in other regions. Additionally, a larger number of pores were found in the lower central section, which may be due to the narrow girdle width. This narrow width can impede the escape of bubbles, causing them to become trapped below the girdle as the molten pool metal solidifies.

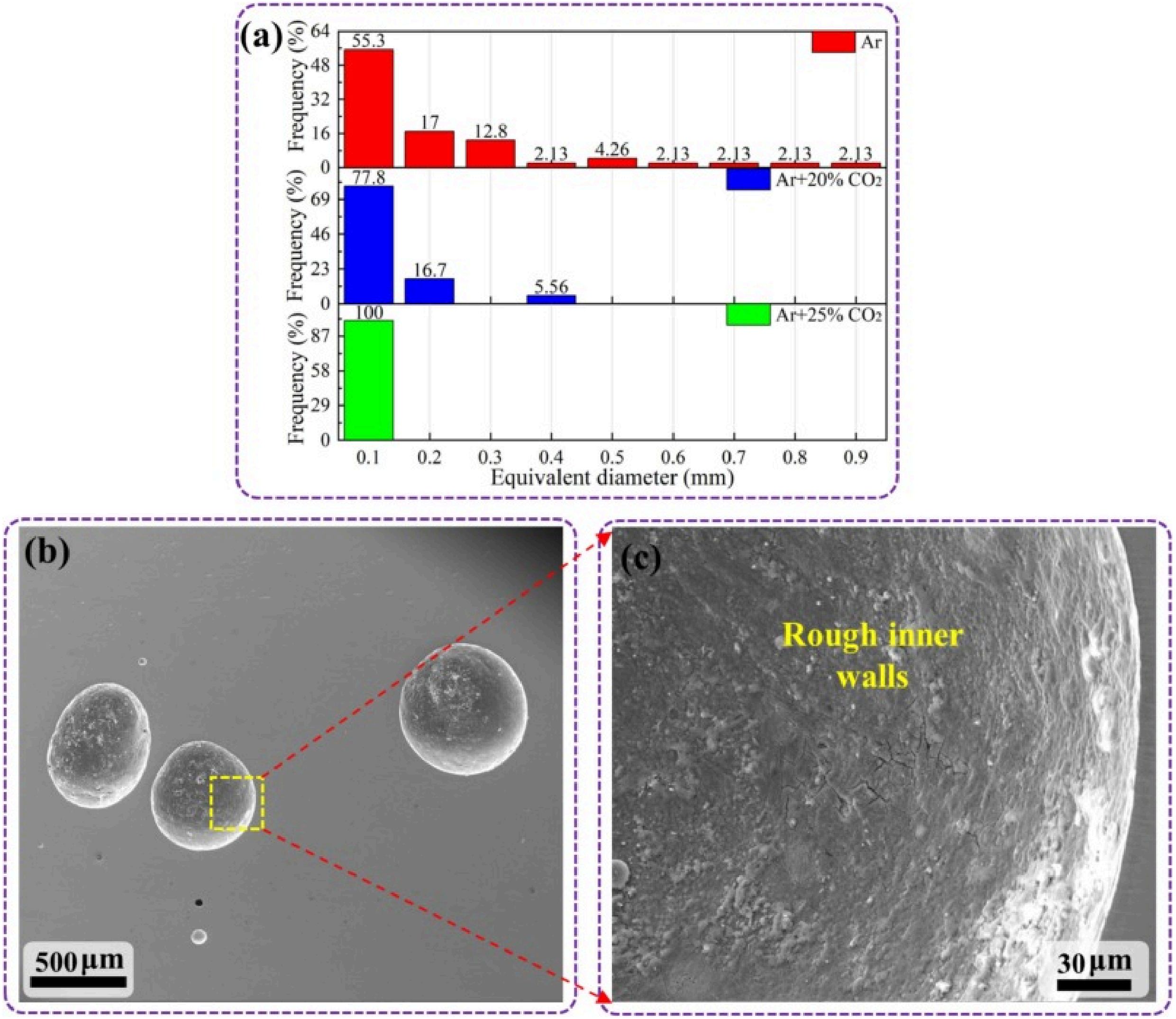


Download: Download high-res image (615KB)

Download: Download full-size image

Fig. 5. Pore distribution measured by μ-CT scanning: (a), (d) and (g) pure Ar; (b), (e) and (h) Ar+20%CO_2; (c), (f) and (i) Ar+25%CO_2.

The pore size and distribution statistics from Fig. 5 are presented in Fig. 6(a). It can be seen that the maximum equivalent pores size reaches 0.9mm, with 27.7% of the pores exceeding 0.2mm in size, under the pure Ar condition. However, when 20%CO_2 is introduced, the maximum pore size is reduced to 0.4mm, with only 5.56% exceeding 0.2mm. Additionally, with the addition of 25%CO_2, the maximum pore size remains below 0.1mm. This suggests that incorporating a specific amount of oxidizing gas into the shielding gas during LCHW of D406A UHSS can significantly minimize or even eliminate welding pores. To further identify the types of pores, the internal morphologies of the pores were characterized. As shown in Fig. 6(b-c), the inner walls of the pores are rough and present an oxidized appearance. This indicates that these pores are typical process pores.



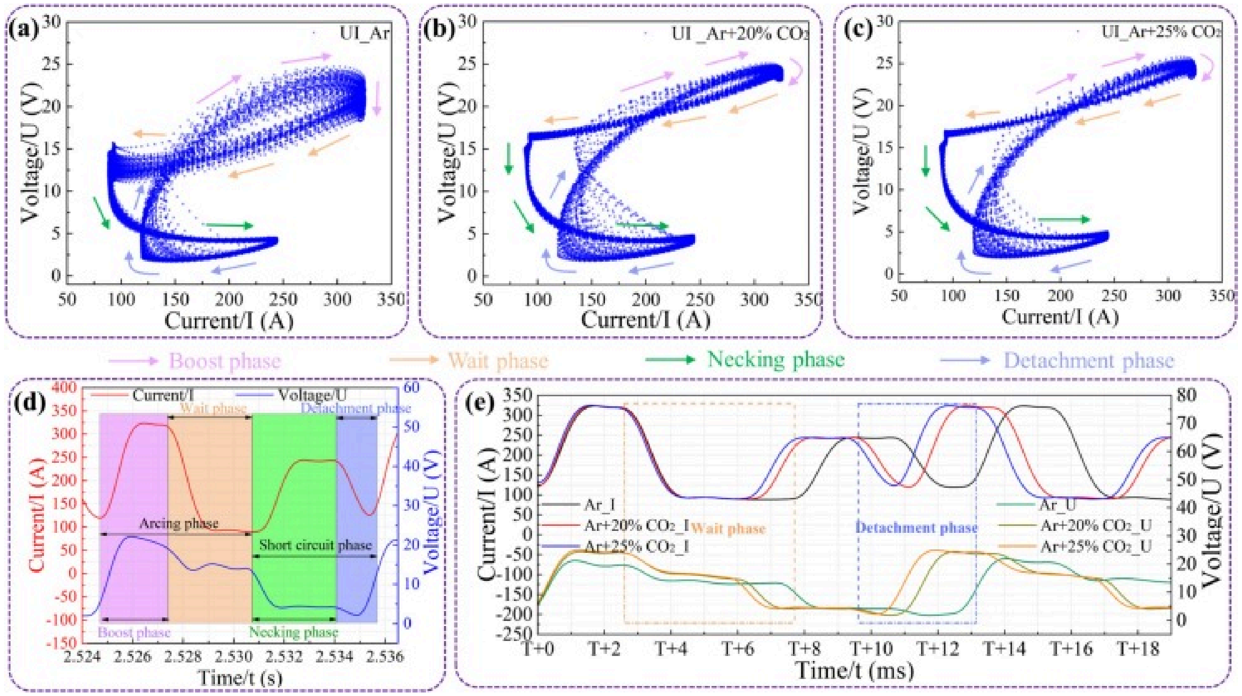
Download: Download high-res image (711KB)

Download: Download full-size image

Fig. 6. Pore size and distribution with different shielding gases (a), the SEM image of pores (b) and the inner wall morphology of pore at high magnification (c).

3.2.2. Effect of different shielding gases on electrical signal

The voltage and current (U/I) diagram is a useful tool for evaluating the stability of the arc during the welding process [36], as the stability of the arc is closely linked to the quality of the weld [37]. In this experiment, real-time welding data was collected, including current and voltage signals, to obtain U/I diagrams under different shielding gases, as shown in Fig. 7(a-c). Based on the actual welding current and voltage waveforms of CMT, one cycle can be divided into four phases: Boost, wait, necking, and detachment [38], as shown in Fig. 7(d). The boost and wait phases are also referred as the arcing phase, while the necking and detachment phases are known as the short circuit phase. Fig. 7(a) displays the U/I diagram for pure Ar, and it can be observed that from detachment to the waiting phase, the U/I diagram is more scattered and occupies a larger area, indicating a larger number of unstable moments during the welding process. Fig. 7(b-c) show the U/I diagrams for mixed gas conditions. It is evident that, apart from the scattered data in the detachment phase, the overall fluctuation level is relatively low. This suggests that with the gradual addition of the CO_2, the unstable moments shift progressively from the boost phase to the detachment phase and gradually decrease.



Download: Download high-res image (764KB)

Download: Download full-size image

Fig. 7. U/I diagrams under different shielding gas: (a) pure Ar, (b) Ar+20%CO_2, (c) Ar+25%CO_2; (d) schematic diagram of the phases in one cycle of CMT welding; (e) comparisons of current and voltage waveforms in one CMT cycle under different shielding gas.

The comparison of current and voltage waveforms during a stable welding process is shown in Fig. 7(e). While the current maintains consistent base and peak values, the duration of each cycle varies depending on the type of shielding gas used. Table 3 presents statistics on the duration of the four phases in one cycle for different CO_2 concentrations. It was found that when pure Ar is used as the shielding gas, the wait and detachment phases are longer. Specifically, with the addition of 20% and 25% CO_2 to the shielding gas, the wait phase duration decreases by 26% and 28%, respectively, and the detachment phase by 25% and 36%. The longer waiting and detachment phases allow for better melting of the wire and promotes droplet growth, resulting in larger droplets when pure Ar is used.

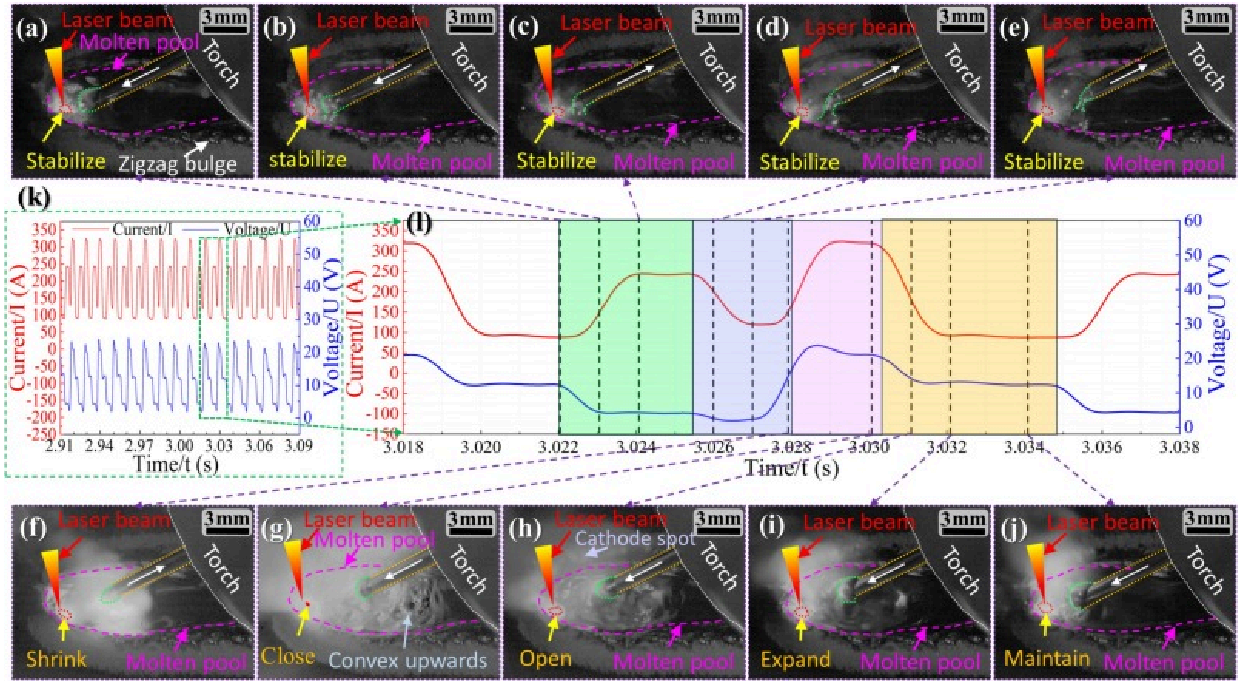
Table 3. Duration of the four phases with different shielding gas.

Shielding gas	Arcing phase		Short circle phase	
	Boost phase (ms)	Wait phase (ms)	Necking phase (ms)	Detachment phase (ms)
Pure Ar	2.4±0.1	**4.6±0.1**	3.4±0.1	**2.3±0.1**
Ar+20%CO_2	2.45±0.1	**3.4±0.1**	3.4±0.1	**1.75±0.1**
Ar+25%CO_2	2.45±0.1	**3.3±0.1**	3.4±0.1	**1.5±0.1**

In summary, as the CO_2 content increases, both the size and number of welding pores decrease significantly due to the stability of the CMT arc. This results in a more stable welding process. Furthermore, the increase in CO_2 content also advances the short-circuit time, promoting the transition of the droplet.

3.3. Droplet transfer behavior under different shielding gases

Fig. 8 shows the droplet transfer behavior with pure Ar shielding gas in one cycle. Initially, the droplet touches the molten pool in an upward position off the axis of the wire, causing the arc to extinguish, as depicted in Fig. 8(a). This indicates that the welding process has entered the necking phase. As the current increases and the voltage decreases, the droplet rapidly constricts due to the wire retraction and a high short-circuit current, forming a liquid bridge, as illustrated in Fig. 8(b-c). When the current decreases further, the welding process enters the detachment phase, where the liquid bridge becomes slender under the influence of surface tension and gravity, as displayed in Fig. 8(d-f). Eventually, the liquid bridge breaks and the metal moves toward the keyhole. During the boost phase, both the current and voltage continuously rise, increasing the electromagnetic force and causing the wire to melt and form a new droplet. Simultaneously, the absence of oxides on the molten pool surface causes the arc to search for cathode spots, resulting in arcs drift [39,40]. These cathode spots have a high current density and arc pressure, while non-cathode spots have a low current density and arc pressure. As a result, the molten pool surface experiences highly uneven flow, leading to various protrusions. This uneven flow also interferes with the stability of the keyhole, causing it to sharply reduce or even collapse, as shown in Fig. 8(g). It is worth noting that the liquid metal solidifies quickly when it reaches the edge of the molten pool, resulting in a zigzag bulge, as seen in Fig. 8(a). As the welding wire advances, the welding process enters a waiting phase, characterized by a decline in the welding current and voltage. This reduces the arc pressure and the extent of protrusions in the molten pool, as illustrated in Fig. 8(h-j). Subsequently, the droplet gradually grows and moves upward off-axis until it contacts the molten pool, marking the beginning of the next cycle. In summary, three behaviors can be observed: (1) The droplet maintains a close distance to the keyhole. (2) The keyhole initially shrinks and may even close as the droplet exits the molten pool, but then gradually expands, and stabilizes. (3) The droplet moves upward off the wire axis with prolonged transit time.

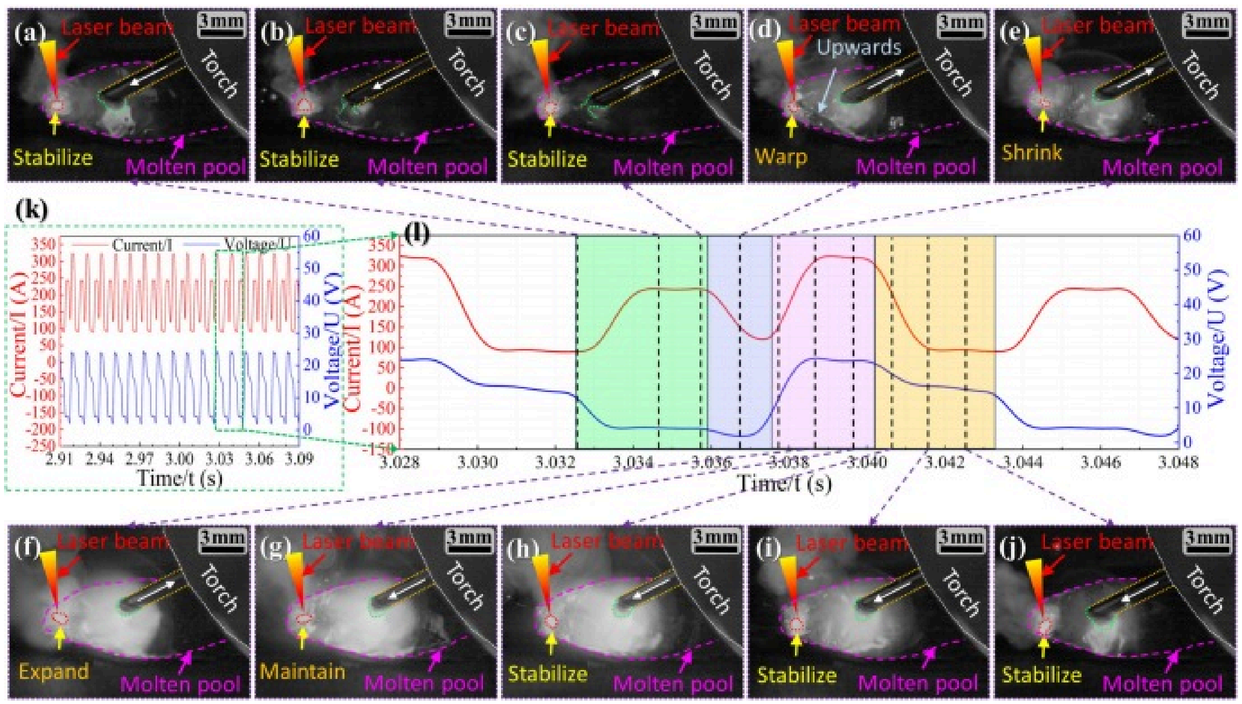


Download: Download high-res image (1MB)

Download: Download full-size image

Fig. 8. Droplet transfer behavior with shielding gas pure Ar: (a-j) droplet behavior at different times, (k) characteristics of electrical signal in several cycles, (l) electrical signal characteristics corresponding to the droplet transfer.

Fig. 9 shows the droplet transfer behavior with Ar+20%CO_2 shielding gas in one cycle. In Fig. 9(a), it can be seen that after the addition of 20% CO_2, the droplet makes contact with the molten pool in a downward position off the axis of the wire, forming an inverted pear shape. During the necking phase, the arc is extinguished, the current increases to 250 A, and the voltage decreases. This leads to a rapid transition of the droplet into the molten pool, as shown in Fig. 9(b-c). As the current decreases, marking the start of the detachment phase, the liquid bridge becomes slender and the time for the liquid bridge fracture is advanced. Simultaneously, the arc quickly reignites, with the impact of the liquid bridge and arc forces driving the molten pool metal toward the keyhole, disrupting its stability, as illustrated in Fig. 9(d). With an increase in the current and voltage, the welding progresses into the boost phase. During this phase, the presence of the active element O in the molten pool ensures stable cathode spots and continuous arc combustion. The droplet continues to grow downward off the axis of the wire, as displayed in Fig. 9(e-h). When the current and voltage decrease, the welding enters the waiting phase, during which the droplet further grows downward off the axis of the wire, gradually approaching the molten pool, as shown in Fig. 9(i-j). In summary, compared with the pure Ar shielding, three distinct behaviors are observed under the Ar+20%CO_2 shielding: (1) The distance between the droplet and the keyhole significantly increases. (2) The shape of the keyhole mouth initially warps as the droplet detaches from the molten pool, then shrinks, expands, and eventually stabilizes. (3) The droplet grows downward away from the axis of the wire, with a noticeably reduced transit time.

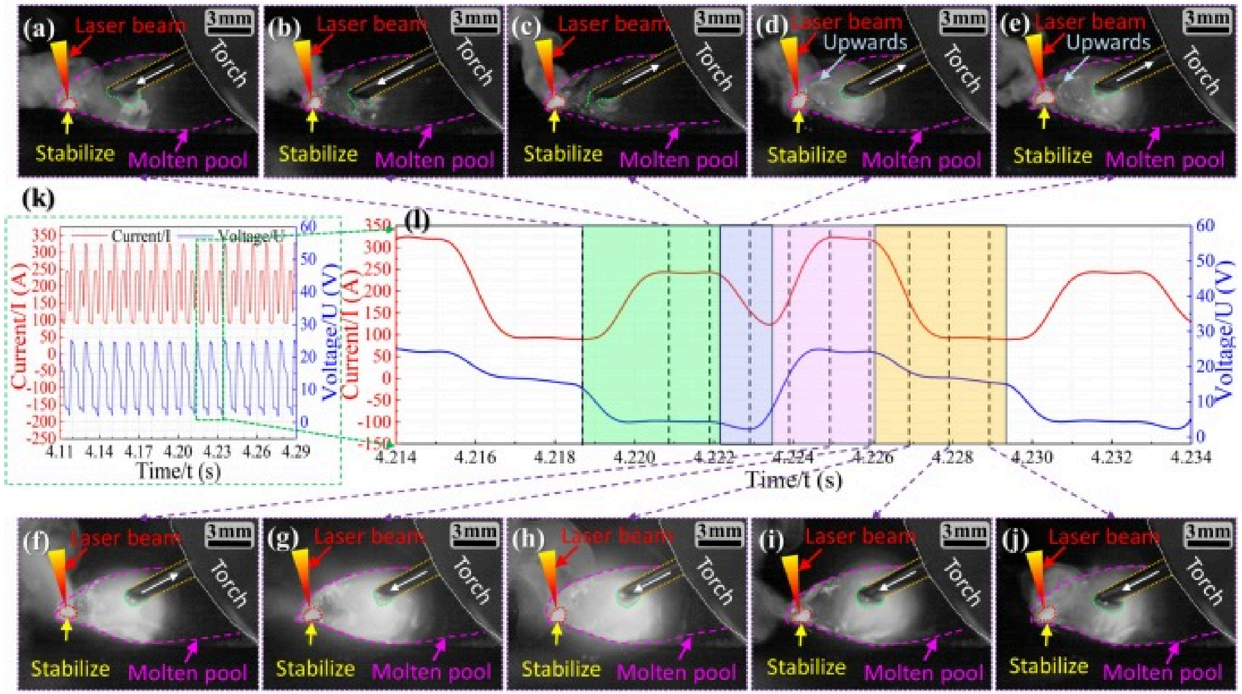


Download: Download high-res image (880KB)

Download: Download full-size image

Fig. 9. Droplet transfer behavior with shielding gas Ar+20%CO_2: (a-j) Droplet behavior at different times, (k) characteristics of electrical signal in several cycles, (l) Electrical signal characteristics corresponding to the droplet transfer.

Fig. 10 shows the droplet transfer behavior with Ar+25%CO_2 shielding gas in one cycle. It was evident that the welding process is more stable and the transition time of the droplet is significantly reduced. In comparison to other shielding gases, the Ar+25%CO_2 condition exhibits three distinct behaviors: (1) The distance between the droplet and the keyhole increases, resulting in a more stable droplet transfer. (2) Despite the metal moving toward the keyhole, the shape of the keyhole mouth remains unchanged after the droplet detaches. (3) The droplet continues to grow downward away from the axis of the wire, leading to a further reduction in transfer time. As shown in Fig. 10(d), during the detachment phase of the welding process, the liquid bridge breaks and the metal moves toward the keyhole. However, unlike in Fig. 9(d), the molten pool metal only accumulates around the keyhole without significantly breaching it.



Download: Download high-res image (886KB)

Download: Download full-size image

Fig. 10. Droplet transfer behavior with shielding gas Ar+25%CO_2: (a-j) Droplet behavior at different times, (k) characteristics of electrical signal in several cycles, (l) Electrical signal characteristics corresponding to the droplet transfer.

Quantitative statistics were conducted to measure the distance (L) between the droplet and the center of the keyhole when it made contact with the molten pool under various shielding gases, as illustrated in Fig. 11. The results revealed that as the CO_2 content increases to 25%, the distance (L) also increases by approximately 2.2 times. This can be attributed to the droplet growing downward off the wire axis due to the presence of oxidizing gas. This downward growth of the droplet can lead to an earlier short circuit at a constant wire feed speed. Therefore, this shift can also be beneficial for maintaining keyhole stability and preventing the collapse of the keyhole.

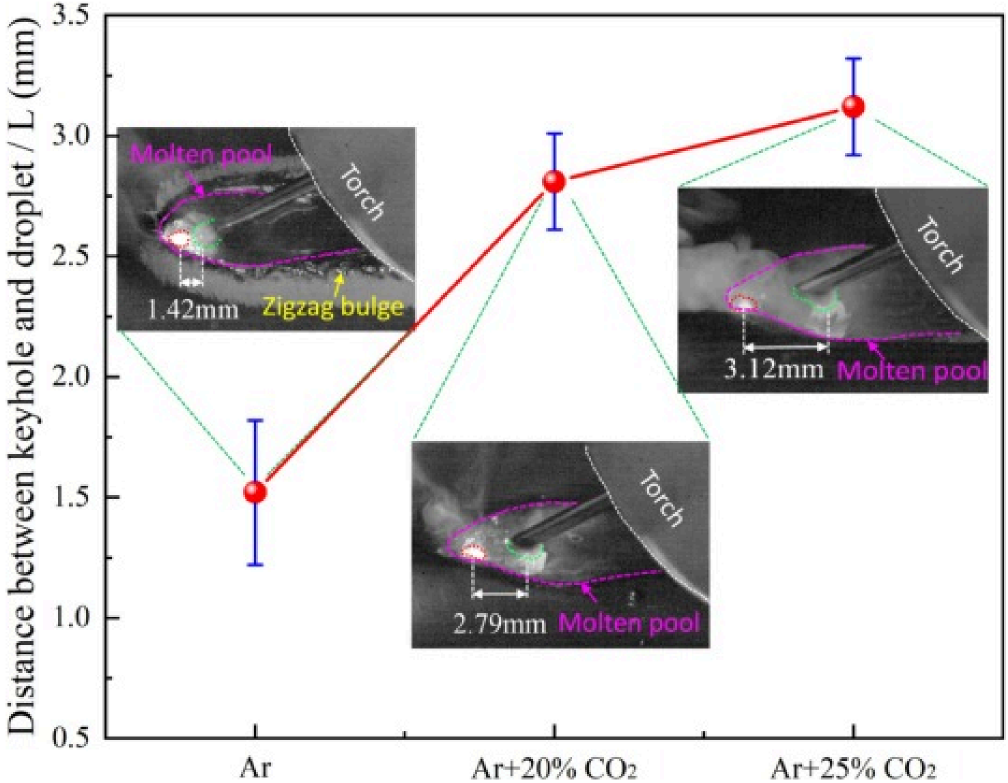
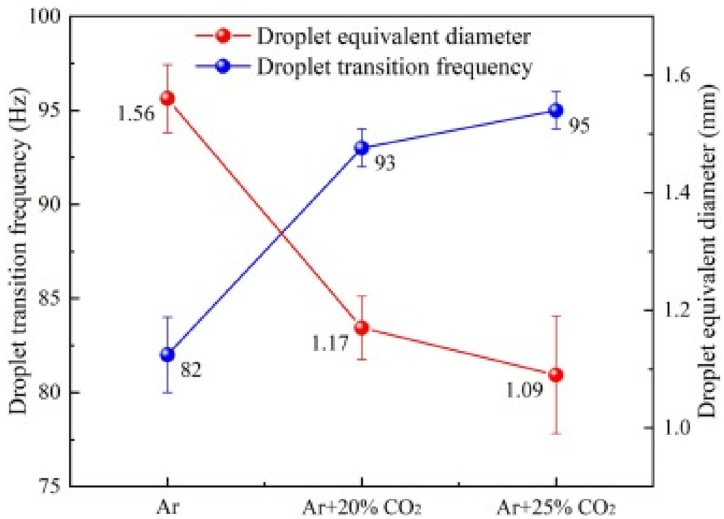


Fig. 11. Distance between keyhole and droplet under different shielding gas.

Fig. 12 shows the variations in droplet transfer frequency and droplet equivalent diameter with different CO_2 contents. According to Table 3, the growth time of the droplet shortens with the addition of CO_2, and the duration of one cycle also progressively decreases. Consequently, with the incorporation of CO_2, the droplet transfer frequency in the LCHW accelerated to 95 Hz, accompanied by a 30.1% reduction in droplet diameter. A smaller droplet diameter enhances the stability of the welding process during transfer, which can further reduce disturbance to the keyhole.

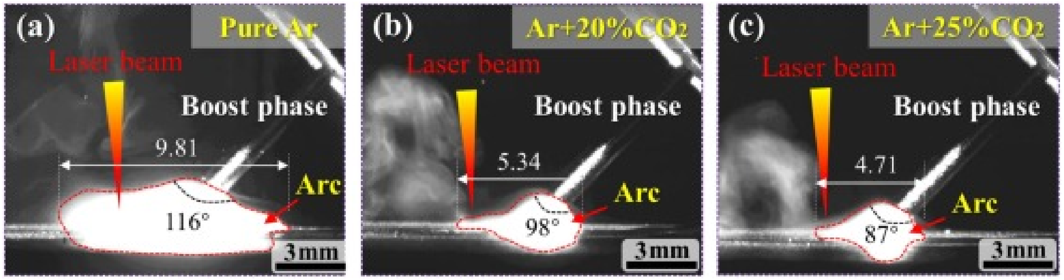


Download: Download high-res image (136KB)

Download: Download full-size image

Fig. 12. Droplet equivalent diameter and transition frequency under different shielding gas.

The arc shape in the boost phase was observed from a vertical plane view, as shown in Fig. 13. The results show that with the increase of CO_2 content, the arc area of CMT decreases gradually, the width of arc area decreases by 52%, and the spreading angle of arc root decreases by 25%. This indicates that CO_2 has the effect of compressing arc, which is consistent with the research results of et al. In addition, the separation of metal vapor induced by laser and arc shows that the laser has little effect on the arc after adding 20% or 25% CO_2.



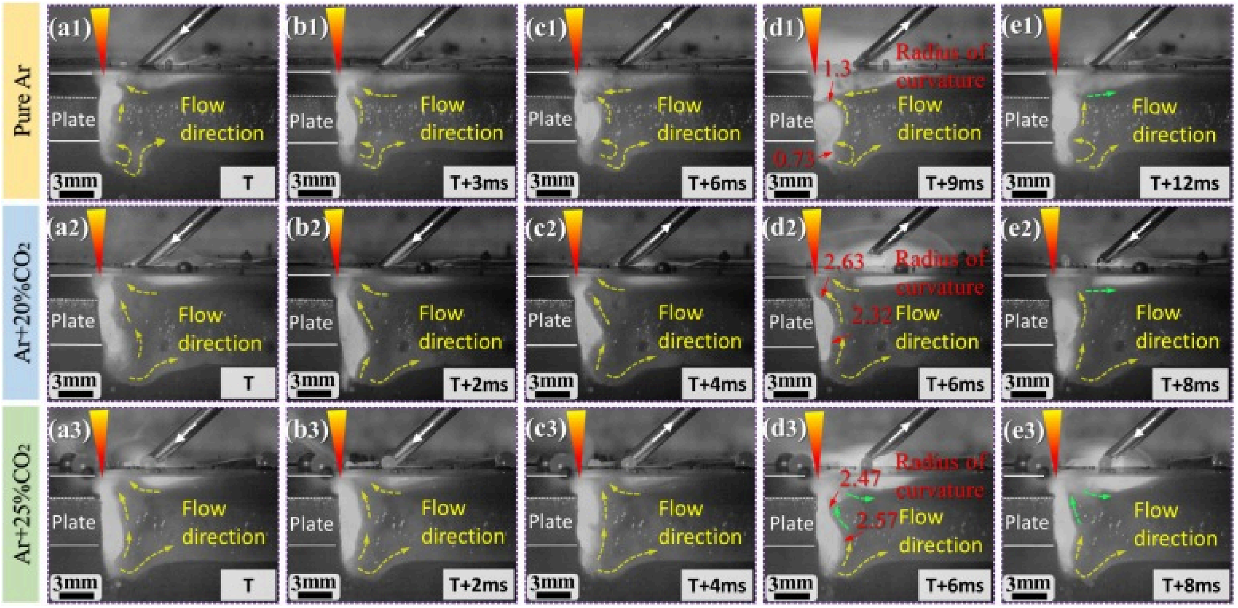
Download: Download high-res image (207KB)

Download: Download full-size image

Fig. 13. Arc and laser plasma behavior: (a) pure Ar, (b)Ar+20%CO_2, (c) Ar+25%CO_2.

3.4. Molten pool internal flow behavior under different shielding gases

Fig. 14 depicts the changes in the flow behavior of molten pool metal under different gas conditions: pure Ar, Ar+20%CO_2, and Ar+25%CO_2. The molten pool metal can be divided into top and bottom sections for analysis, based on its flow patterns in the thickness direction.



Download: Download high-res image (860KB)

Download: Download full-size image

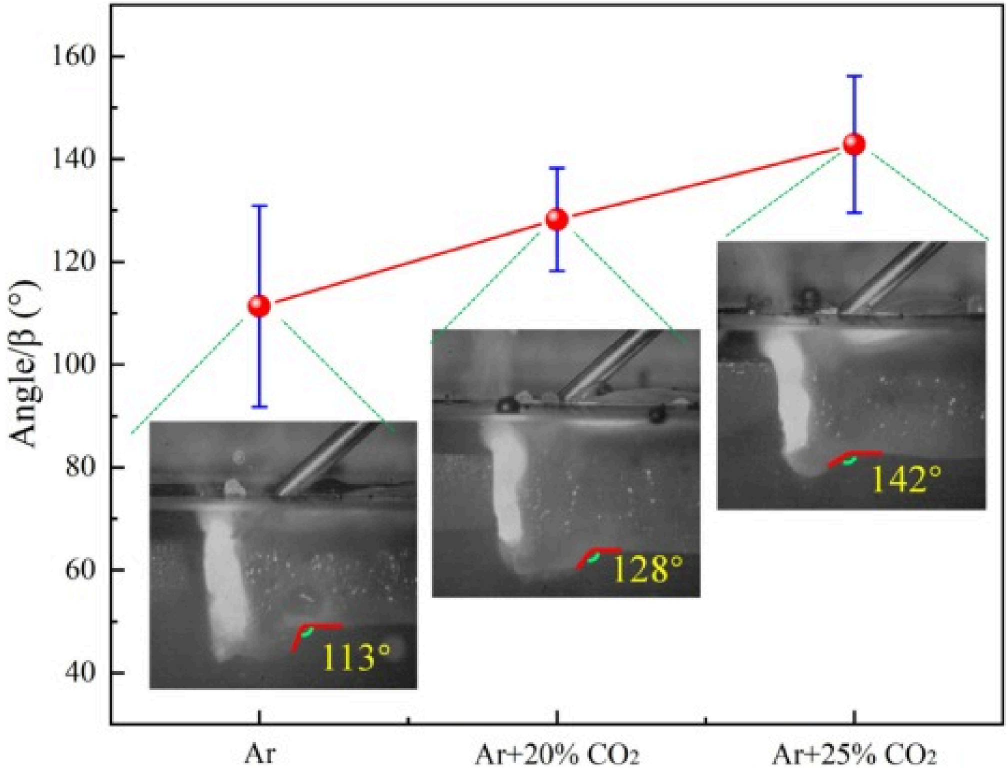
Fig. 14. Molten pool flow behavior in vertical plane view with transparent quartz glass: (a1-e1) pure Ar, (a2-e2) Ar+20%CO_2, (a3-e3) Ar+25%CO_2.

As shown in Fig. 14(a1-e1), the top liquid metal experiences violent fluctuations due to the entry of metal droplets under the pure Ar shielding. When the wire is fed, the top liquid metal flows toward the keyhole and is also pushed upwards by the vapor recoil force. As the wire is retracted, the liquid bridge breaks, increasing the driving force for the liquid metal to flow toward the keyhole. This results in a counterclockwise flow, as shown in Fig. 14(d1), which overcomes the vapor recoil force and fully covers the keyhole. As the welding current decreases, the driving force toward the keyhole also decreases, causing the top liquid metal to move away from the keyhole due to vapor recoil pressure. Contrastingly, the bottom liquid metal at the rear wall of the keyhole flows backward along the molten pool boundary and exhibits a counterclockwise flow, resulting in a distinct bulge.

As shown in Fig. 14(a2-e2), the top liquid metal also exhibits a counterclockwise flow toward the keyhole under an Ar+20%CO_2 shielding. However, although the keyhole mouth shrinks significantly during wire retraction, it is not entirely covered by liquid metal. As the welding progresses, the vapor recoil force dominates the flow, directing the metal toward the tail of the molten pool. This reduces the counterclockwise flow at the bottom of the keyhole rear wall, resulting in a decrease in the bulge.

Fig. 14(a3-e3) also show the flow behavior of the molten pool under an Ar+25%CO_2 shielding. In contrast to pure Ar and Ar+20%CO_2, the top liquid metal flows clockwise, as depicted in Fig. 14(d3). This aids in keyhole expansion and prevents defects, such as pores, resulting from keyhole collapse.

In addition to changes in the keyhole shape and metal flow at the keyhole rear wall with different shielding gases, significant differences in the flow angle of the bottom metal were also observed. The findings are summarized in Fig. 15. It was observed that as the CO_2 content increases from 0 to 25%, the flow angle of the metal at the bottom of the molten pool increases linearly. A smaller angle resulted in a more pronounced counterclockwise flow of metal at the bottom of the molten pool, which tended to close the keyhole at the bottom. This impeded bubble escape and exacerbated the formation of pores. Fortunately, as the angle increases, the metal at the bottom of the molten pool flows away from the keyhole, resulting in a smoother flow that facilitated bubble escape.



Download: Download high-res image (205KB)

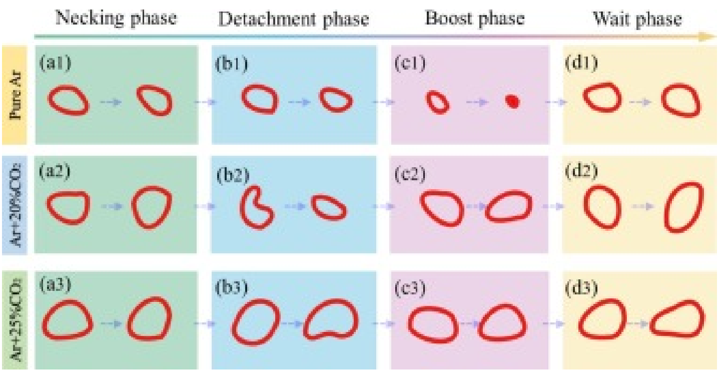
Download: Download full-size image

Fig. 15. Bottom liquid metal flow angle under different shielding gas.

3.5. Mechanisms of pore suppression based on the dynamic of the droplet transfer process and keyhole stability

3.5.1. Keyhole mouth evolution

As the CO_2 content increases, droplet short-circuiting and detachment occurred earlier, as shown in Table 3. The evolution of the keyhole mouth shape is closely linked to the droplet detachment process. Hence, a schematic diagram (Fig. 16) of the shape evolution for keyhole mouth was obtained by extracting and enlarging in the same scale in Fig. 8, Fig. 9, Fig. 10.



Download: Download high-res image (133KB)

Download: Download full-size image

Fig. 16. Evolution of keyhole mouth shape under different shielding gas: (a1-d1) pure Ar, (a2-d2) Ar+20%CO_2, (a3-d3) Ar+25%CO_2.

Under the pure Ar shielding, the liquid bridge breaks during the late detachment phase, while the droplet fully enters the molten pool during the boost phase. This results in the keyhole mouth shrinking and potentially collapsing due to the high current and large droplets. However, as the droplet transition is complete, the keyhole gradually reopens and stabilizes, as shown in Fig. 16(d1).

When using Ar+20%CO_2, the droplet transfer frequency increases, causing the liquid bridge to break earlier in the detachment phase. This leads to the keyhole warping and the molten pool flowing during this phase. As the droplet detaches, the keyhole mouth shrinks, as shown in Fig. 16(b2). But then gradually expands and recovers during the boost phase, as illustrated in Fig. 16(c2-d2).

Under the Ar+25%CO_2 shielding, the liquid bridge also breaks early in the detachment phase, but the smallest droplet size and increased *L* mitigate the disruption caused by the droplet impact on the keyhole rear wall. Consequently, the keyhole mouth remains relatively stable, as depicted in Fig. 16(a3-d3). In conclusion, a reduction or even collapse of the keyhole mouth can prevent the internal metal vapor from escaping, leading to an increase in the porosity, as demonstrated in Fig. 3 and Fig. 5. As the CO_2 content increases, the keyhole mouth stabilizes and tends to enlarge.

3.5.2. Dynamic analysis of the droplet transfer process

Differences in droplet morphology and transition frequency across various welding phases and shielding gases can disrupt keyhole stability, potentially resulting in defects such as porosity. Moreover, the morphology and transition frequency of droplets are also closely related to the forces during the welding process. Based on droplet transfer studies, the effects of gravity F_G, surface tension F_s, electromagnetic force F_e, plasma flow force F_P, and metal vapor jet force F_r on droplets and liquid bridges in LCHW are individually considered. Details of these effects are outlined below.

1) The gravity of droplet (F_G) [41]:

$$\mathrm{F_G} = \frac{4}{3}\pi \mathrm{r_d^3}\rho \mathrm{g} \tag{1}$$

In Eq. (1), r_d is the droplet radius, ρ is the density of the droplet, and g is the acceleration of gravity. Without considering the influence of other factors, it can be approximated that $F_G \propto r_d^3$. Therefore, based on the high-speed camera images and the measurements in Fig. 12, the droplet gravity ratio under the shielding gas of pure Ar, Ar+20%CO_2 and Ar+25%CO_2 were approximately: 1: 0.43: 0.34.

2) The surface tension of droplet (F_s) [42,43]:

$$\mathrm{F_s} = 2\pi \mathrm{r_w f}\gamma \tag{2}$$

$$\gamma = \gamma_m^o - A(T - T_m) - RT\Gamma_s\, ln\left[1 + k_1\alpha_o \exp\left(-\frac{\Delta H_0}{RT}\right)\right] \tag{3}$$

In Eqs. (2), (3), r_w is the radius of filler wire, f is the correction factor of the empirical formula, γ is the surface tension of melting metal, A is the surface tension coefficient (4.3 $\times 10^{-4}$ $\mathrm{N/(m \bullet K)}$), T_m is the melting temperature of pure metal (1812K), T is the fluid temperature, γ_m^o is the surface tension of pure metal at melting temperature (1.943 $\mathrm{N/m}$), R is the gas constant (8.3143 $\mathrm{J/(mol \bullet K)}$), Γ_s is the surface excess at saturation(2.03 $\times 10^{-8}$ $\mathrm{mol/m^2}$), k_1 is a constant (1.38×10^{-2}), α_o is the thermodynamic activity of surface-active component oxygen (wt%O), and ΔH_0 is the standard heat of absorption (-1.46×10^5 $\mathrm{J/mol}$).

The oxygen content of weld metals under pure Ar, Ar+20%CO_2 and Ar+25%CO_2 shielding was measured by LECO ONH836 equipment, which were 0.002%, 0.00809% and 0.0157%, respectively. From Eqs. (2), (3), $F_s \propto \gamma$ can be approximately considered. At T=1683K (melting point), the surface tensions were calculated to be 1.8075 $\mathrm{N/m}$, 1.5484 $\mathrm{N/m}$ and 1.3898 $\mathrm{N/m}$ for pure Ar, Ar+20%CO_2 and Ar+25%CO_2 shielding, respectively. In other words, the ratio of the surface tension applied to the molten droplets were approximated to be 1: 0.86: 0.77.

3) The electromagnetic force on droplet (F_e) [12]:

$$\mathrm{F_e} = \frac{\mu_0 \mathrm{I}^2}{4\pi}\left[\ln\left(\frac{\mathrm{r_d}\sin\theta}{\mathrm{r_w}}\right) - \frac{1}{1-\cos\theta} + \frac{2}{(1-\cos\theta)^2}\ln\left(\frac{2}{1+\cos\theta}\right) - \frac{1}{4}\right] \tag{4}$$

In Eq. (4), I is the welding current and θ is the droplet conductive zone half cone angle, respectively. μ_0 is vacuum magnetic permeability ($4\pi \times 10^{-7}$ $\mathrm{H/m}$). From Eq. (4) F_e is not only proportional to I^2 but also closely related to θ. It has been shown that when θ <30°, the direction of the electromagnetic force changes from positive to negative [44], indicating a shift from a detaching force to a retaining. In addition, since the current values are almost equal at each phase, the shielding gas has an approximately equal effect on the electromagnetic force. That is to say, the ratio of the electromagnetic force at each phase with pure Ar, Ar+20%CO_2, and Ar+25%CO_2 shielding was approximately: 1: -1: −1.

In particular, the electromagnetic force is minimized at the waiting phase due to the lowest current.

4) The plasma flow force (F_P) [45]:

$$F_P = \frac{1}{2} C_0 \rho_g v_g^2 \pi r_d^2 \left(1 - \frac{r_w^2}{2r_d^2}\right) \tag{5}$$

In Eq. (5), C_0 is the drag coefficient of metal vapor, ρ_g and v_g are the density and velocity of metal vapor, respectively. Without considering the influence of other factors, it can be approximated that $F_P \propto r_d^2\left(1 - \frac{r_w^2}{2r_d^2}\right)$. Thus, based on the droplet diameter measurements in Fig. 12, the plasma flow force ratios for pure Ar, Ar+20%CO_2 and Ar+25%CO_2 shielding can be obtained to be approximately: 1: 0.38: 0.27.

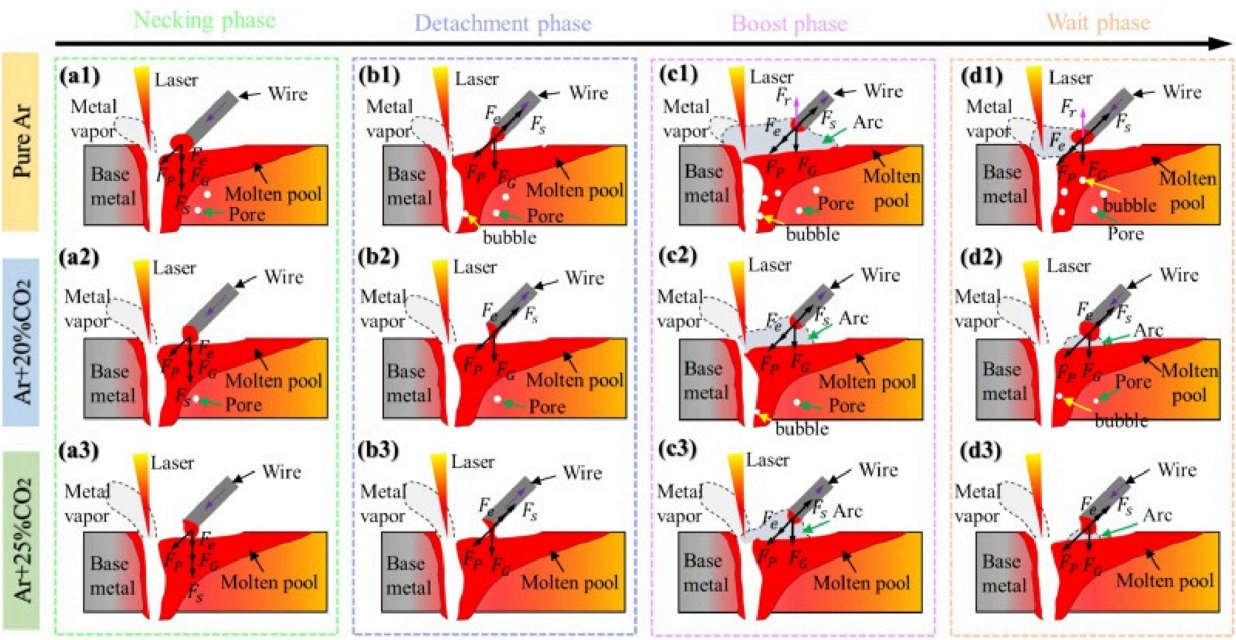
5) Metal vapor jet force (F_r) [46]:

$$F_r = \begin{cases} \frac{1}{4\pi R_h^2} C_0 A_0 \rho_m^2 V_0^2 \left(\frac{N_A k_B T_s^{3/2}}{M_a B_0}\right) \exp\left(-\frac{U}{T_s}\right) \exp\left(-\frac{L^2}{2R_h^2}\right), (L \le R_h) \\ 0, (L > R_h) \end{cases} \tag{6}$$

In Eq. (6), R_h represents the maximum radius of metal vapor, A_0 is the area of plasma acting, ρ_m is the density of plasma, V_0 is the fixed value, T_s is the temperature of melted metal, N_A is Avogadro's constant, k_B is Boltzmann constant, and L is the distance between laser keyhole and droplet. From Fig. 13, the maximum radius R_h of the metal vapor protected by pure Ar, Ar+20%CO_2 and Ar+25%CO_2 were approximated to be 4.905 mm, 2.67 mm and 2.355 mm, respectively. Meanwhile, from Fig. 11, the distances L between the keyhole and the molten droplet are approximated to be 1.42 mm, 2.79 mm and 3.12 mm, respectively. From Eq. (6), it can be seen that the metal vapor jet force occurs only in the case of pure Ar shielding while it was 0 in the case of Ar+20%CO_2 and Ar+25%CO_2 shielding. Therefore, it can be considered that the ratio of the metal vapor jet force in the case of pure Ar, Ar+20%CO_2, and Ar+25%CO_2 protection was approximately: 1: 0: 0.

In summary, with the increase of CO_2 content, the order of the change of each force is as follows: $F_r > F_e > F_P > F_G > F_s$.

Fig. 17 illustrates the droplet morphology and force schematics during the four phases of LCHW under three different shielding gases. In Fig. 17(a1-a3), the droplet morphology and force balances during the necking phase are shown under various shielding gases. During this phase, gravity, electromagnetic forces, plasma flow forces and surface tension all contribute to a smooth transition of the droplets into the melt pool.



Download: Download high-res image (692KB)

Download: Download full-size image

Fig. 17. Schematic diagram of droplet transfer behavior and force analysis on different phases with shielding gas: (a1-d1) pure Ar, (a2-d2) Ar+20%CO_2, (a3-d3) Ar+25%CO_2.

Fig. 17(b1-b3) show the droplet morphology and force balances during the detachment phase under different shielding gases. At this time, the liquid bridge experiences significant necking and eventually breaks. As the welding wire retracts, the electromagnetic force and surface tension acting on the metal of the liquid bridge near the welding wire are directed upward. With the addition of CO_2, the surface tension, gravity, and plasma flow force acting on the droplet decrease. However, due to the varying degrees of influence of each force, the resultant force is directed upward. Therefore, the duration of this phase is continuously shortened with the increase of CO_2. Additionally, if the components of the electromagnetic force, surface tension, and plasma flow force in the welding direction exceed the force stabilizing the keyhole, the keyhole collapse and bobble are prone to occur.

Fig. 17(c1-c3) show the droplet morphology and force balances during the boost phase under various shielding gases. During this phase, the droplets under pure Ar shielding are additionally subjected to the metal vapor jet force, which encourages the droplets to move upward away from the wire axis. However, as the CO_2 content increases to 20% or 25%, the metal vapor jet force disappears, gravity decreases, while the plasma flow force decreases to a greater extent than the surface tension, so the droplet has a tendency to grow behind the wire axis. In addition, the intense laser-arc interaction under pure Ar caused fluctuations in the molten pool, which disturbed the stabilization of the keyhole and led to a further increase in gas bubbles. With the increase of CO_2, the stability of the keyhole gradually increased and the number of bubbles gradually decreased.

Fig. 17(d1-d3) show the droplet morphology and force balances during the wait phase under various shielding gases. As the filler wire advances, the distance between keyhole and droplet progressively decreases and the vapor jet force becomes stronger. This causes the droplet to move upwards, away from the axis of the wire when pure Ar is used for shielding. However, when the amount of CO_2 increases, the angle of droplet conductive zone decreases (as shown in Fig. 8(j), Fig. 9(j) and Fig. 10(j)), resulting in a change of the electromagnetic force from a detached force to a retaining force. Additionally, the resultant force of surface tension, gravity and plasma flow force are upwards. Therefore, the droplet to move downwards away from the axis of the wire. The more the droplet deviates from the axis, the sooner it makes contact with the molten pool, leading to a short circuit and shortening the welding cycle. This also increases the transition frequency.

3.5.3. Flow dynamics in the keyhole rear wall and pore suppression

As the CO_2 content increases, the plasma composition within the arc space changes, subsequently impacting the flow of the molten pool and the stability of the keyhole. Research indicated that the horizontal pressure balance on the keyhole rear wall is primarily maintained by the supersaturated vapor pressure ΔP_g, surface tension pressure P_s, arc pressure P_a, and hydrostatic pressure P_h [33,47]. In addition, according to Fig. 8(e), Fig. 9(c) and Fig. 10(c), it can be seen that the heights of the molten droplet to the molten pool were approximate and the sizes of the liquid bridge were small, so the impacts of the liquid bridge on the keyhole were not considered in this paper. To determine the reason for CO_2 improving the keyhole stability in LCHW, an analysis was conducted on the pressure balance in the keyhole rear wall with pure Ar, Ar+20%CO_2, and Ar+25%CO_2 shielding gases. Furthermore, because of the keyhole is responsive to the horizontal force component, this study focuses on the horizontal force on the keyhole rear wall.

As mentioned above, the pressure balance equation of the liquid metal on the keyhole rear wall in the horizontal direction can be expressed as:

$$\Delta \mathrm{P_g} = \mathrm{P_s} + \mathrm{P_h} + \mathrm{P_a} \tag{7}$$

where,

$$\Delta \mathrm{P_g} = \mathrm{P_v} - \mathrm{P_0} \tag{8}$$

In this formula, P_v represents the metal vapor pressure in the keyhole and P_0 denotes the ambient pressure. Thus, the pressure balance equation for the metal at the rear wall of the keyhole under ideal conditions can be described as:

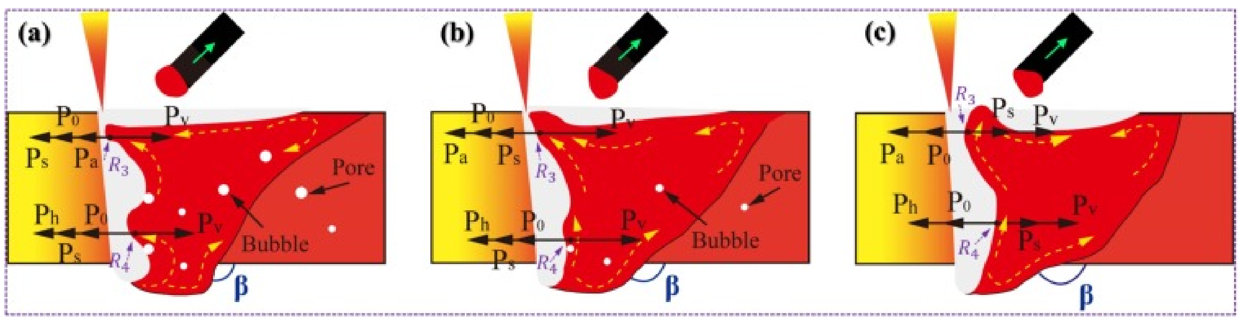
$$P_s + P_0 + P_a + P_h = P_v \tag{9}$$

In the equation, the force on the left promotes keyhole closure, while the force on the right promotes keyhole opening. Since the variation of P_0, P_h and P_v are small, the effect of shielding gas variation on these forces were not considered in this paper. However, with the change of shielding gas, the surface tension pressure and arc pressure at the rear wall of the keyhole produce significant changes, so these two forces are discussed separately:

Surface tension pressure (P_s) [48,49]:

$$\mathrm{P_s} = \gamma\left(\frac{1}{\mathrm{R_3}} + \frac{1}{\mathrm{R_4}}\right) \tag{10}$$

where, γ is the surface tension of the molten metal and R_3 and R_4 are the principal radii of the surface curvature (Fig. 18). From the actual radius of curvature measured in Fig. 14(d1-d3) and Eq. (3), the surface tension pressures under the shielding of pure Ar, Ar+20%CO_2 and Ar+25%CO_2 were calculated to be 3.87 Kpa, 1.26 Kpa and 1.1 Kpa, respectively.



Download: Download high-res image (239KB)

Download: Download full-size image

Fig. 18. Schematic diagram of molten pool flow and keyhole pattern under different shielding gas: (a) pure Ar, (b) Ar+20%CO_2, (c) Ar+25%CO_2.

Arc pressure (P_a) [50]:

$$P_a = \frac{\mu_0 I^2}{8\pi r_a} \exp\left(-\frac{L^2}{2r_a^2}\right) \tag{11}$$

where, r_a is the effective radius of the arc and L is the distance between droplet and keyhole. According to Fig. 13, the r_a under different shielding gases were 4.905 mm, 2.67 mm and 2.355 mm, respectively. Hence, combining with L in Fig. 11, the arc pressures at the rear wall of the keyhole under the shielding of pure Ar, Ar+20%CO_2 and Ar+25%CO_2 can be calculated to be 0.21 Kpa, 0.43 Kpa and 0.43 Kpa, respectively.

Fig. 18 shows the schematic diagram of molten pool flow and keyhole pressure balance with different shielding gas on the boost phase. Under the pure Ar shielding, the liquid metal surface at the rear wall of the keyhole is rough and the curvature change is small. Calculations show that this surface tension pressure was high and the metal vapor pressure was not enough to maintain the keyhole open, resulting in the collapse of the keyhole. Further, the collapsed keyhole prevents the escape of high temperature vapors from the keyhole, resulting in the generation of bubbles, as shown in Fig. 18(a). Meanwhile, the metal at the top of the rear wall of the keyhole flows toward the keyhole under the arc pressure. However, the metal at the back of the molten pool was driven to flow clockwise by the Marangoni flow, which further makes it difficult for the gas bubbles to escape from the molten pool. Additionally, the metal at the bottom of the keyhole rear wall flows toward the keyhole due to the bottom angle β, which increases the surface tension pressure and accelerates the protrusion formation. This further leads to a large number of porosities in the weld metal.

Under the Ar+20%CO_2 shielding, the liquid metal surface of the keyhole rear wall is relatively smooth and the radius of curvature increases. Calculations show that the surface tension pressure decreased from 3.87 Kpa to 1.26 Kpa, while the arc pressure of the metal at the keyhole rear wall increased from 0.21 Kpa to 0.43 Kpa. This suggests that the pressure preventing the formation of the keyhole decreases, which contributes to the stabilization of the keyhole. However, the metal vapor pressure is still insufficient to maintain the stability of the keyhole, leading to the generation of small bubbles, as shown in Fig. 18(b). Meanwhile, the addition of oxidizing gases causes a shift in the Marangoni force of the metal behind the molten pool from clockwise to counterclockwise flow [51], which can help the bubbles to escape and reduce the number of pores in the weld metal. In addition, the increase of the bottom angle β also makes the counterclockwise flow tendency of the metal decrease, which can inhibit the formation of the protrusion of the rear wall of the keyhole and further reduce the generation of gas bubbles.

Under the Ar+25%CO_2 shielding, the liquid metal surface of the rear wall of the keyhole is further smoothed with a larger and stable radius of curvature. Based on the shape of the rear wall of the keyhole, it can be inferred that the direction of the surface tension pressure is toward the right at this time, which promotes the opening of the keyhole. The arc pressure, which prevents the formation of the keyhole, decreases from 0.43 Kpa to 0.4 Kpa due to the large distance from the keyhole. As a result, the metal flow pattern at the top keyhole rear wall changes from counterclockwise to clockwise, as shown in Fig. 18(c). Under the effect of metal vapor pressure, the keyhole further opens up, which prevents the keyhole from collapsing. The Marangoni flow of the metal behind the molten pool further enhances the counterclockwise flow of the molten pool and reduces the probability of the appearance of porosity in the weld. In addition, the increase of the bottom angle β further inhibited the formation of the protrusion. These factors prevent keyhole collapse and avoid the generation of pores in the weld metal.

4. Conclusions

In this paper, the weld formation, pore distribution, and electrical signal characteristics were studied, and the droplet transfer as well as keyhole rear wall liquid metal flow behavior were also explored under different shielding gases. Furthermore, the mechanisms behind keyhole stabilization and weld porosity were fully revealed in LCHW of UHSS. This work provides an effective approach for suppressing weld porosity in UHSS and lays a strong foundation for the future application of LCHW in the aerospace industry. The following conclusions were drawn:

1. When the CO_2 content in the shielding gas was increased from 0 to 30%, the porosity of the LCHW weld gradually decreased until it disappeared. Initially, the weld morphology improved, but then it worsened. The optimal weld was achieved when the shielding gas consisted of Ar and 25% CO_2, with no porosity present.

2. The transition frequency increased from 82 Hz to 95 Hz when the CO_2 content increased to 25%. Conversely, the diameter of the droplet decreased from 1.56 mm to 1.09 mm. These changes can reduce the impact of the molten droplet on the keyhole. Moreover, the shape of the droplet shifted from a downward inverted pear-shaped to an upward spherical shape, leading to an increase in the distance between the droplet and the keyhole from 1.42 mm to 3.12 mm. This can help to keep the keyhole open and provide a favorable channel for bubble escape.

3. The direction of molten metal flow on the top of the keyhole rear wall changed from counterclockwise to clockwise when the CO_2 content increased to 25%. As a result, the previously unstable collapse and warp of the keyhole mouth gradually shifted to a stable opening. Furthermore, the alteration in flow dynamics behavior of the molten pool has caused an increase in the bottom flow angle from 113° to 142°. Accordingly, the counterclockwise flow trend of metal in the keyhole rear wall gradually decreased. This maintains the stability of the keyhole internal shape, contributing to reducing the likelihood of pore formation.

4. The metal vapor jet force, the plasma flow force and surface tension gradually decrease, promoting the transfer of droplets with the CO_2 content increases. While during the waiting phase, the variation of electromagnetic

force from a detached force to a retaining force leads to an increase in the distance between the keyhole and droplet, which further weakens the impact of vapor jet force on droplets. Meanwhile, the surface tension pressure gradually decreases from 3.86 Kpa to 1.1 Kpa, and at Ar+25% CO_2, itpromote the opening of keyhole and further suppress the formation of the protrusion on the keyhole rear wall. These factors worked together to improve the stability of the keyhole and suppress the formation of pores.

CRediT authorship contribution statement

Yuhui Zhang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cong Chen:** Visualization, Validation, Methodology, Investigation, Formal analysis. **Changjian Wang:** Validation, Resources. **Ran Xiong:** Supervision, Resources. **Ke Zhang:** Visualization, Supervision, Resources, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

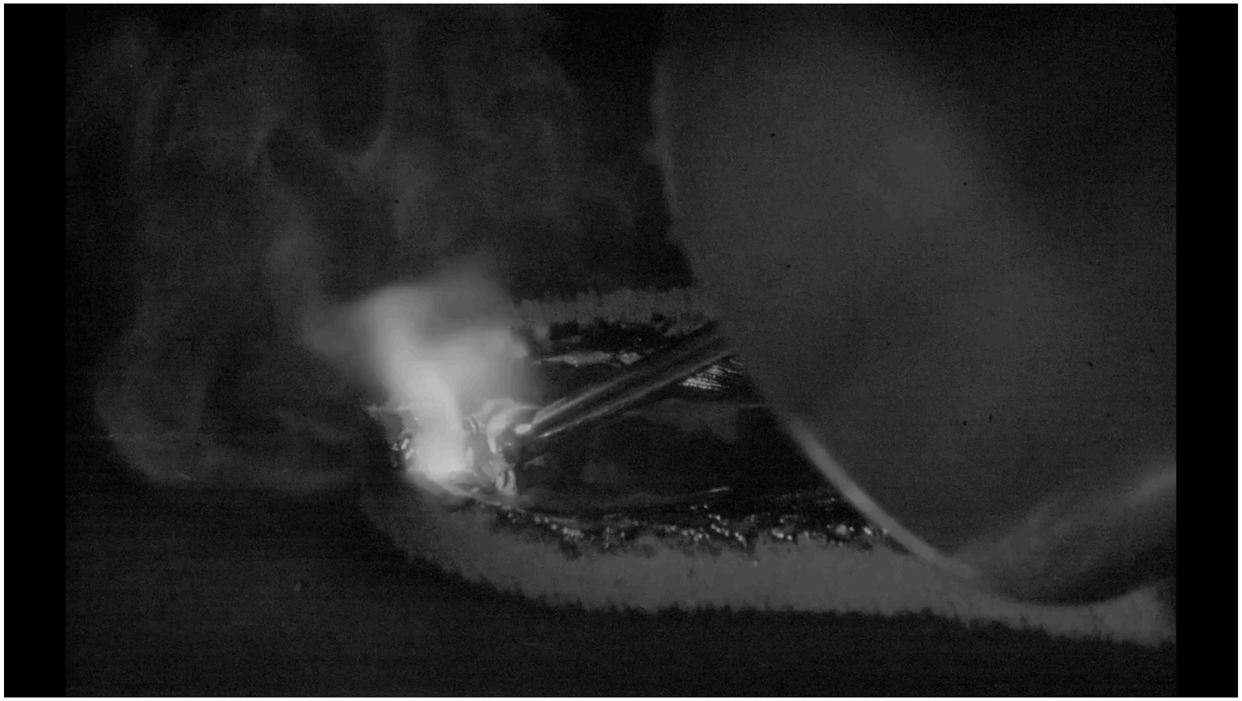
Acknowledgements

This research is financially supported by the National Natural Science Foundation of China [grant numbers 51875353 and 51875354].

Appendix A. Supplementary data

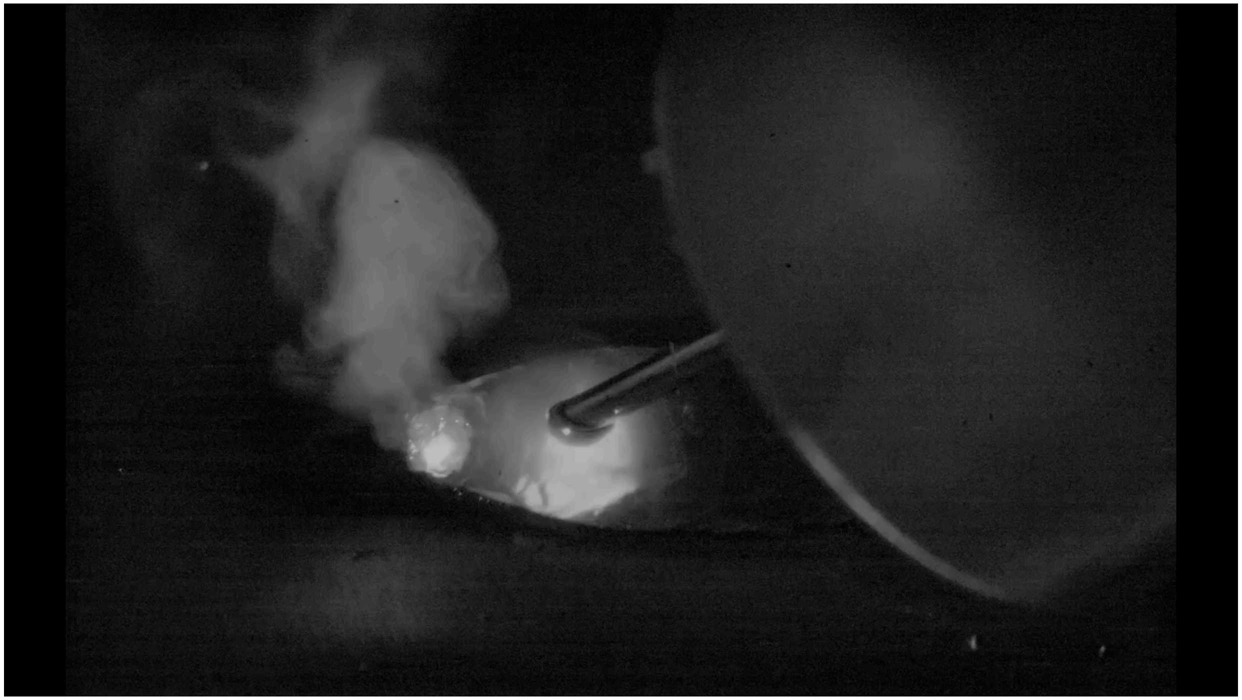
Download all supplementary files included with this article What's this?

The following are the supplementary data related to this article.



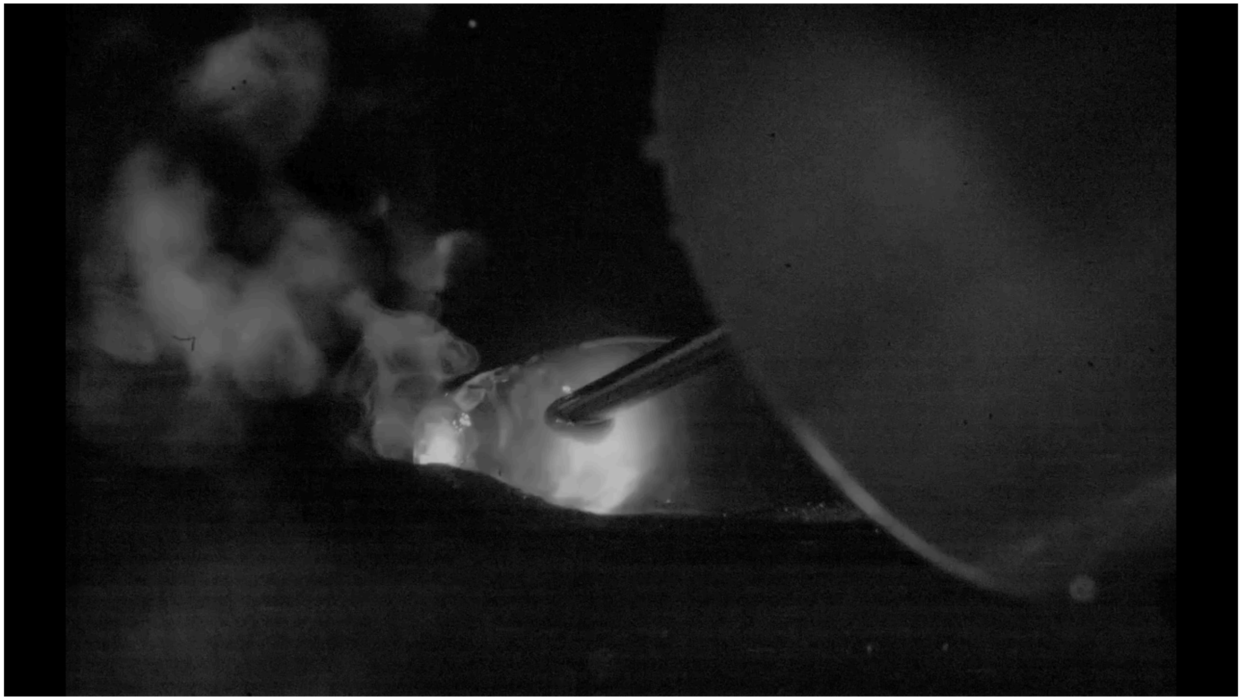
Download: Download video (5MB)

Supplementary video 1. Droplet transfer behavior with pure Ar.



Download: Download video (5MB)

Supplementary video 2. Droplet transfer behavior with Ar + 20%CO_2.

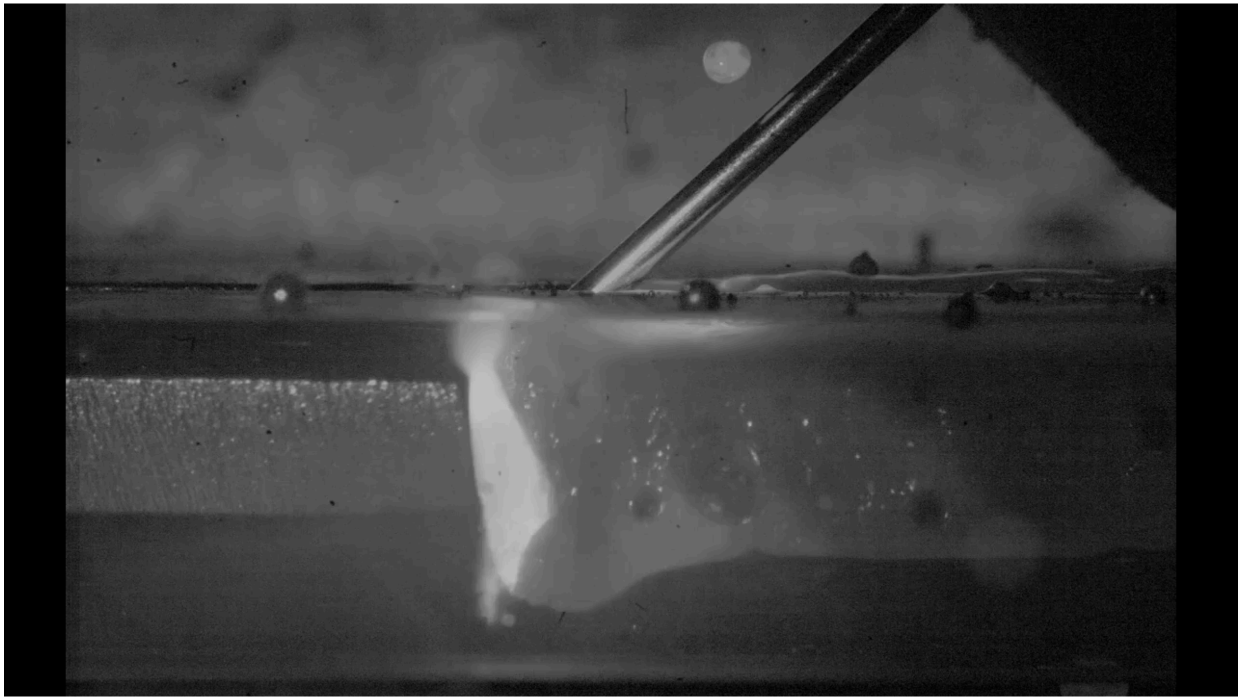


Download: Download video (5MB)

Supplementary video 3. Droplet transfer behavior with Ar + 25%CO_2.

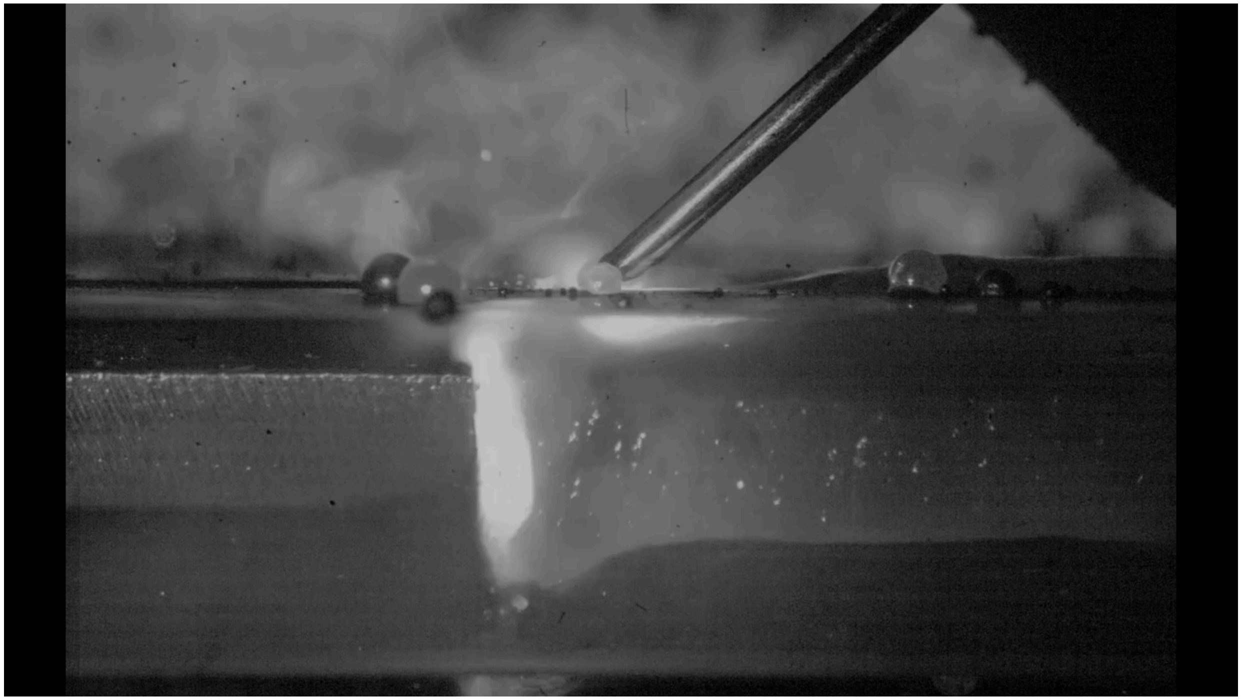
Download: Download video (2MB)

Supplementary video 4. Molten pool internal flow behavior with pure Ar.



Download: Download video (2MB)

Supplementary video 5. Molten pool internal flow behavior with Ar + 20%CO_2.



Download: Download video (2MB)

Supplementary video 6. Molten pool internal flow behavior with Ar + 25%CO_2.

References

[1] C. Chen, W. Wang, D. Li, Y. Cai, Y. Zhang, K. Zhang
Effect of beam oscillation on microstructure and properties of laser-TIG hybrid welding of D406A ultra-high strength steel
J Manuf Process, 57 (2020), pp. 798-805
View PDF View article View in Scopus ↗ Google Scholar ↗

[2] Z. Liu, X. Li, L. Pan, J. Gao, K. Zhang
Effects of weld penetration modes on laser welding characteristics of a novel ultra-high strength steel for aerospace application
J Manuf Process, 90 (2023), pp. 111-124

View PDF View article Google Scholar ↗

[3] J. Li, D. Zhan, Z. Jiang, H. Zhang, Y. Yang, Y. Zhang
Progress on improving strength-toughness of ultra-high strength martensitic steels for aerospace applications: a review
J Mater Res Technol, 23 (2023), pp. 172-190
View PDF View article Google Scholar ↗

[4] H. Zhu, Z. Xiong, J. Mao, X. Cheng
Effect of ageing temperature on the microstructural evolution and mechanical properties in M2C and NiAl co-precipitation secondary hardening ultrahigh-strength steel
J Mater Res Technol, 30 (2024), pp. 3522-3534
View PDF View article View in Scopus ↗ Google Scholar ↗

[5] H. Wang, D. Liu, J. Wang, Y. Yang, H. Wang, N. Lv, *et al.*
Simultaneous increase of ultimate tensile strength and uniform elongation of 30Si2MnCrMoVE ultra-high strength steel by hot deformation direct quenching and partitioning
J Mater Res Technol, 19 (2022), pp. 404-417
View PDF View article Crossref ↗ View in Scopus ↗ Google Scholar ↗

[6] M. St, M. Zeman Węglowski
Prevention of cold cracking in ultra-high strength steel Weldox 1300
Archives of Civil and Mechanical Engineering, 14 (2014), pp. 417-424
Google Scholar ↗

[7] C. Chen, H. Zhou, C. Wang, L. Liu, Y. Zhang, K. Zhang
Laser welding of ultra-high strength steel with different oscillating modes
J Manuf Process, 68 (2021), pp. 761-769
View PDF View article View in Scopus ↗ Google Scholar ↗

[8] M. Xie, Z. Lv, J. Wu, L. Zhang, H. Yu, J. Long, *et al.*
Laser butt welding of 6.6-mm thick D406A ultrahigh strength steel: part I —porosity and lower surface depression
J Mater Eng Performance (2024), 10.1007/s11665-024-09903-8
Google Scholar

[9] J. Ning, L.-J. Zhang, J.-n. Yang, X.-Q. Yin, X.-W. Wang, J. Wu, Characteristics of multi-pass narrow-gap laser welding of D406A ultra-high strength steel, J Mater Process Technol, 270 (2019) 168–181.
Google Scholar

[10] P.C. Adamczuk, I.G. Machado, J.A.E. Mazzaferro
Methodology for predicting the angular distortion in multi-pass butt-joint welding
J Mater Process Technol, 240 (2017), pp. 305-313
View PDF View article View in Scopus Google Scholar

[11] B. Acherjee
Hybrid laser arc welding: state-of-art review
Optics & Laser Technology, 99 (2018), pp. 60-71
View PDF View article View in Scopus Google Scholar

[12] H. Huang, P. Zhang, H. Yan, Z. Liu, Z. Yu, D. Wu, *et al.*
Research on weld formation mechanism of laser-MIG arc hybrid welding with butt gap
Optics & Laser Technology, 133 (2021)
Google Scholar

[13] S. Ma, X. Chen, M. Jiang, B. Li, Z. Wang, Z. Lei, *et al.*
Surface morphology, microstructure and mechanical properties of Al–Mg–Sc alloy thin wall produced by laser-arc hybrid additive

manufacturing
Thin-Walled Structures, 186 (2023)
Google Scholar ↗

[14] M. Zhang, Z. Zhang, L. Lei, W. Zhou, M. Du, B. Zhang
Effect of repair weld length on microstructure evolution and mechanical properties of 30CrMnSiNi2A ultra-high strength steel
J Mater Res Technol, 23 (2023), pp. 4828-4842
View PDF View article View in Scopus ↗ Google Scholar ↗

[15] S. Zhang, Y. Zhu, F. Zhang, X. Guo, Y. Xu, H. Wang, *et al.*
Novel flat-top laser-aided cold metal transfer additive manufacturing for titanium alloy: arc characteristics, microstructure, and tensile properties
J Mater Process Technol, 327 (2024)
Google Scholar ↗

[16] J. Frostevarg, A.F.H. Kaplan, J. Lamas
Comparison of CMT with other arc modes for laser-arc hybrid welding of steel
Welding in the World, 58 (2014), pp. 649-660
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[17] Z. Lei, B. Li, J. Bi, P. Zhu, W. Lu, J. Liu
Influence of the laser thermal effect on the droplet transfer behavior in laser-CMT welding
Optics & Laser Technology, 120 (2019)
Google Scholar ↗

[18] G. Li, C. Zhang, M. Gao, X. Zeng
Role of arc mode in laser-metal active gas arc hybrid welding of mild steel
Materials & Design, 61 (2014), pp. 239-250
View PDF View article View in Scopus ↗ Google Scholar ↗

[19] Q. Liu, D. Wu, Q. Wang, P. Zhang, H. Yan, T. Sun, *et al.*
Progress and perspectives of joints defects of laser-arc hybrid welding: a review
Int J Adv Manuf Technol, 130 (2023), pp. 111-146
Google Scholar ↗

[20] C. Zhao, N.D. Parab, X. Li, K. Fezzaa, W. Tan, A.D. Rollett, *et al.*
Critical instability at moving keyhole tip generates porosity in laser melting
Science, 370 (2020), pp. 1080-1086
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[21] C. Chen, C. Wang, N. Chen, L. Liu, W. Wang, K. Zhang
Effect of multiple hit and break of oscillating pulse laser-TIG hybrid welding on porosity suppression of ultra-high strength steel
J Manuf Process, 83 (2022), pp. 387-397
View PDF View article Crossref ↗ View in Scopus ↗ Google Scholar ↗

[22] M. Mazar Atabaki, J. Ma, W. Liu, R. Kovacevic
Pore formation and its mitigation during hybrid laser/arc welding of advanced high strength steel
Materials & Design, 67 (2015), pp. 509-521
View PDF View article View in Scopus ↗ Google Scholar ↗

[23] C. Jiang, X. Li, H. Luo, J. Li, S. Qu, C. Yang
DT300 ultra-high strength steel fabricated using selective laser melting: densification, microstructure, and mechanical properties
Mater Sci Eng A, 863 (2023)
Google Scholar ↗

[24] G. Tani, G. Campana, A. Fortunato, A. Ascari
The influence of shielding gas in hybrid LASER–MIG welding

Appl Surf Sci, 253 (2007), pp. 8050-8053

View PDF View article View in Scopus ↗ Google Scholar ↗

[25] Q. Liu, D. Wu, Q. Wang, P. Zhang, H. Yan, T. Sun, *et al.*
Research status of stability in dynamic process of laser-arc hybrid welding based on droplet transfer behavior: a review
Coatings, 13 (2023)

Google Scholar ↗

[26] K.M. Hafez
The influence of shielding gases on keyhole-induced porosity and nitrogen absorption in SS 304 stainless steel fiber laser welds
The International Journal of Advanced Manufacturing Technology, 127 (2023), pp. 1887-1894

Crossref ↗ View in Scopus ↗ Google Scholar ↗

[27] Y. Zhu, Y. Cai, M. Wang
Effects of He content in shielding gases on high-efficient hybrid laser arc welding with C-276 filler metal
J Mater Process Technol, 299 (2022)

Google Scholar ↗

[28] A. Fellman, V. Kujanpää
The effect of shielding gas composition on welding performance and weld properties in hybrid CO2 laser–gas metal arc welding of carbon manganese steel
J Laser Appl, 18 (2006), pp. 12-20

Crossref ↗ View in Scopus ↗ Google Scholar ↗

[29] C. Cai, S. He, H. Chen, W. Zhang
The influences of Ar-He shielding gas mixture on welding characteristics of fiber laser-MIG hybrid welding of aluminum alloy
Optics & Laser Technology, 113 (2019), pp. 37-45

View PDF View article Google Scholar ↗

[30] S. Zhang, J. Sun, M. Zhu, L. Zhang, P. Nie, Z. Li
Effects of shielding gases on process stability of 10CrNi3MoV steel in hybrid laser-arc welding
J Mater Process Technol, 270 (2019), pp. 37-46
View PDF View article Crossref ↗ Google Scholar ↗

[31] Q. Pan, M. Mizutani, Y. Kawahito, S. Katayama
Effect of shielding gas on laser–MAG arc hybrid welding results of thick high-tensile-strength steel plates
Welding in the World, 60 (2016), pp. 653-664
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[32] M. Wahba, M. Mizutani, S. Katayama
Hybrid welding with fiber laser and CO2 gas shielded arc
J Mater Process Technol, 221 (2015), pp. 146-153
View PDF View article View in Scopus ↗ Google Scholar ↗

[33] L. Zhao, S. Tsukamoto, G. Arakane, T. Sugino
Prevention of porosity by oxygen addition in fibre laser and fibre laser–GMA hybrid welding
Science and Technology of Welding and Joining, 19 (2014), pp. 91-97
View in Scopus ↗ Google Scholar ↗

[34] S. Lu, H. Fujii, K. Nogi
Marangoni convection and weld shape variations in Ar–O2 and Ar–CO2 shielded GTA welding
Mater Sci Eng A, 380 (2004), pp. 290-297
View PDF View article View in Scopus ↗ Google Scholar ↗

[35] X. Cui, J. Chen, M. Chen, C. Wu

Investigating arc and molten metal transport phenomena in gas metal arc welding with Ar–CO2 gas mixtures using a numerical method
Phys Fluids, 36 (2024)
Google Scholar ↗

[36] K. Wu, M. Zhang, Y. Zeng, X. Hong, P. Xie
Evaluation of arc stability in double-wire DP-GMAW of aluminum alloy based on U-I graph
J Manuf Sci Eng, 144 (2022)
Google Scholar ↗

[37] S. Zhang, Y. Wang, M. Zhu, Y. Feng, P. Nie, Z. Li
Effects of heat source arrangements on laser-MAG hybrid welding characteristics and defect formation mechanism of 10CrNi3MoV steel
Journal of Manufacturing Processes, 58 (2020), pp. 563-573
View PDF View article View in Scopus ↗ Google Scholar ↗

[38] M. Chen, D. Zhang, C. Wu
Current waveform effects on CMT welding of mild steel
J Mater Process Technol, 243 (2017), pp. 395-404
View PDF View article View in Scopus ↗ Google Scholar ↗

[39] P.J. Modenesi, J.H. Nixon
Arc instability phenomena in GMA welding
Welding Journal (Miami, Fla), 73 (1994), pp. 2195-2245
View in Scopus ↗ Google Scholar ↗

[40] L.J. da Silva, F.M. Scotti, D.B. Fernandes, R.P. Reis, A. Scotti
Effect of O2 content in argon-based shielding gas on arc wandering in WAAM of aluminum thin walls
CIRP Journal of Manufacturing Science and Technology, 32 (2021), pp. 338-345
View PDF View article View in Scopus ↗ Google Scholar ↗

[41] M.S. Kang, H. Chung
Dynamic force balance model considering tapering effect in gas metal arc welding
J Mater Process Technol, 257 (2018), pp. 79-87
View PDF View article View in Scopus ↗ Google Scholar ↗

[42] G. Tang, X. Zhao, R. Li, Y. Liang, Y. Jiang, H. Chen
The effect of arc position on laser-arc hybrid welding of 12-mm-thick high strength bainitic steel
Optics & Laser Technology, 121 (2020)
Google Scholar ↗

[43] J. Xu, P. Zou, W. Wang, D. Kang
Study on the mechanism of surface topography evolution in melting and transition regimes of laser polishing
Optics & Laser Technology, 139 (2021)
Google Scholar ↗

[44] B.Y.-S. Kim, T.W. Eagar
Analysis of metal transfer in gas metal arc welding
Welding Journal-New York, 72 (1993), pp. 269s-278s
Google Scholar ↗

[45] F. Liu, B. Xu, K. Song, C. Tan, H. Zhao, G. Wang, *et al.*
Improvement of penetration ability of heat source for 316 stainless steel welds produced by alternating magnetic field assisted laser-MIG hybrid welding
J Mater Process Technol, 299 (2022)
Google Scholar ↗

[46] Y. Li, S. Geng, Z. Zhu, Y. Wang, G. Mi, P. Jiang

Effects of heat source configuration on the welding process and joint formation in ultra-high power laser-MAG hybrid welding
Journal of Manufacturing Processes, 77 (2022), pp. 40-53
View PDF View article Crossref ↗ Google Scholar ↗

[47] M. Jiang, X. Chen, Y. Chen, W. Tao
Increasing keyhole stability of fiber laser welding under reduced ambient pressure
J Mater Process Technol, 268 (2019), pp. 213-222
View PDF View article View in Scopus ↗ Google Scholar ↗

[48] W.-I. Cho, P. Woizeschke
Analysis of molten pool behavior with buttonhole formation in laser keyhole welding of sheet metal
International Journal of Heat and Mass Transfer, 152 (2020)
Google Scholar ↗

[49] L. Zhao, S. Tsukamoto, G. Arakane, T. Sugino, T. DebRoy
Influence of oxygen on weld geometry in fibre laser and fibre laser–GMA hybrid welding
Science and Technology of Welding and Joining, 16 (2011), pp. 166-173
Google Scholar ↗

[50] W. Liao, Y. Liu, M. Ma, D. Wang, L. Zhou, Z. Wang, *et al.*
Dynamic behaviors and homogenization mechanism of molten pool in laser-arc hybrid welding with beam oscillation
Optics & Laser Technology, 178 (2024)
Google Scholar ↗

[51] A. Van Nguyen, S. Tashiro, M.H. Ngo, A.H. Le, H. Van Bui, M. Tanaka

Influence of shielding gas composition on molten metal flow behavior during plasma keyhole arc welding process

Journal of Manufacturing Processes, 53 (2020), pp. 431-437

View PDF View article View in Scopus ↗ Google Scholar ↗

View Abstract

Recommended articles

Based on reading popularity

Research article Full text access

Microstructure and mechanical properties of laser-welded dissimilar DP780 and DP980 high-strength steel joints

Shuling Gao, ..., Wencong Qiu

Materials Science and Engineering: A • Volume 720 • 2018 • Pages 117-129

View PDF

Research article Open access

Hardness, grainsize and porosity formation prediction on the Laser Metal Deposition of AISI 304 stainless steel

Jon Iñaki Arrizubieta, ..., Amaia Alberdi

International Journal of Machine Tools and Manufacture • Volume 135 • 2018 • Pages 53-64

View PDF

Research article Full text access

Microstructure and dynamic tensile characteristics of dissimilar fiber laser welded advanced high strength steels

M.H. Razmpoosh, ..., Y. Zhou

Materials Science and Engineering: A • Volume 773 • 2020 • Article 138729

View PDF

Research article Full text access

An investigation on rivet plug laser welding of hybrid joints between high-strength steel and aluminum alloy

Zhaoguo He, ..., Jinshui Liu

Optics & Laser Technology • Volume 164 • 2023 • Article 109470

View PDF

Research article Open access

Studying the strengthening mechanisms and mechanical properties of dissimilar laser-welded butt joints of medium-Mn stainless steel and automotive high-strength carbon steel

Atef Hamada, ..., Antti Järvenpää

Materials Science and Engineering: A • Volume 856 • 2022 • Article 143936

View PDF

Research article Full text access

Effect of microstructure on tensile and fatigue properties of laser-welded DP780 and DP980 high-strength steel joints

Lijun Yang, ..., Yuntao Li

Materials Science and Engineering: A • Volume 754 • 2019 • Pages 535-555

View PDF

Cited by (5)

Research article

Shielding gas induced thermodynamics mechanisms in keyhole plasma arc welding studied by experiment and numerical simulation

Wang L., ..., Liu J.

International Journal of Thermal Sciences • Volume 219 • 2026 • Article 110244

Show abstract

Research article

Enhancing deposition rate and mechanical properties through hybrid arc welding with activated flux: A novel approach

Tathgir S., ..., Batish A.

Journal of Manufacturing Processes • Volume 151 • 2025

Show abstract

Research article Open access

Understanding the influence of different HLAW parameters to increase the welding speed of 8 mm thick structural steel

Sánchez-Amaya J.M., ..., Gómez-Parra Á.

Journal of Materials Research and Technology • Volume 39 • 2025

Show abstract

Influence of Different Shielded Gases on Welding Performance in SRA EGW ↗

Chen Y., ..., Fang C.

Metallography Microstructure and Analysis • Volume 15 • 2026

Welding of Steel with a High Carbon Equivalent and Bainite Microstructure with Extremely Low Heat Input ↗

Brykov M.N., ..., Molochkov D.

Construction Materials • Volume 6 • 2026 • Article 9